Journal of International Economics 151 (2024) 103970

Contents lists available at ScienceDirect

Journal of International Economics

journal homepage: www.elsevier.com/locate/jie

Full length articles

US trade policy and the US dollar☆

Makram Khalil *, Felix Strobel

Deutsche Bundesbank, Germany

ARTICLE INFO

Dataset link: Replication package for 'US trade policy and the US dollar' (Original data)

JEL classification:
F31
F13
F14
F41
E31

Keywords:
Trade policy uncertainty
Safe asset currency
Two-country model with financial frictions

ABSTRACT

We show that the US dollar response to trade policy uncertainty (TPU) is key to assessing the impact of US trade policy. Employing structural vector autoregressive models, we find that TPU shocks supported a multilateral USD appreciation during the 2018–19 trade tensions between the US and some of its major trading partners. We rationalize this in a two-country New Keynesian model with financial frictions that links the increase in TPU to rising demand for safe USD assets. Our findings suggest that the TPU-induced appreciation of the USD in 2018–19 significantly counteracted US trade policy attempts to raise US competitiveness.

1. Introduction

In 2018 and 2019, the US administration took a markedly tighter trade policy stance against some of its major trading partners. Most prominently, duties on imports from China were raised substantially in several steps. Such policies typically aim at improving the competitiveness of US-produced goods relative to rest-of-the world products. However, 2018 and 2019 also saw a notable appreciation of the US dollar against a broad set of currencies. The appreciation of the US dollar, in turn, created an opportunity for foreign exporters to lower their prices for exports to the US, thereby improving the competitive position of US trading partners. Against this background, the question arises whether the observed USD appreciation in 2018 and 2019 was driven by changes in US trade policy itself.

We argue that trade conflicts cannot be characterized exclusively by the increase in tariff rates. US trade policy in 2018 and 2019, for instance, was shaped by a continuous process of negotiations and disputes with major trading partners. For market participants, it became increasingly difficult to infer the possible future path of policy actions. Desired or not, US trade policy in these episodes was accompanied by elevated levels of uncertainty.[1] Caldara et al. (2020) construct a trade policy uncertainty (TPU) index based on the relative percentage share of TPU-related content in leading US newspapers and find marked increases in TPU, in particular

☆ We would like to thank the editor, our anonymous referees, Dario Caldara, Peter Egger, Almira Enders, Christoph Fischer, Pierre-Olivier Gourinchas, Johannes Hoffmann, Matthias Hoffmann, Sören Karau, Johanna Krenz, as well as participants at the 2021 conference of the Verein für Socialpolitik, the 14th FIW research conference "International Economics", the 2022 Computing in Economics and Finance Conference, the 2022 International Association for Applied Econometrics Annual Conference, and at seminars at the University of Hamburg, de Nederlandsche Bank, the European Central Bank, and the Deutsche Bundesbank for useful comments. The opinions expressed in this paper are those of the authors and do not necessarily reflect the views of the Deutsche Bundesbank or the Eurosystem.

* Correspondence to: Wilhelm-Epstein Strasse 14, 60431 Frankfurt am Main, Germany.

E-mail addresses: makram.khalil@bundesbank.de (M. Khalil), felix.strobel@bundesbank.de (F. Strobel).

[1] Uncertainty about the evolution of policy measures might not be merely the outcome of policy actions but also itself a policy tool (cf. inter alia Ghironi and Ozhan, 2020, in a monetary policy context).

https://doi.org/10.1016/j.jinteco.2024.103970
Received 14 January 2022; Received in revised form 6 July 2024; Accepted 6 July 2024
Available online 14 July 2024

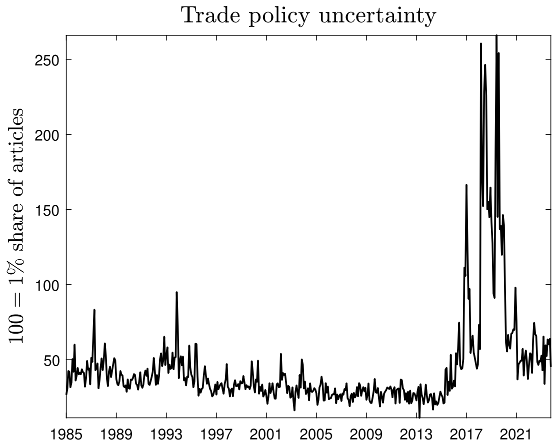

Fig. 1. Trade policy uncertainty index based on Anglo-American newspaper articles. Share of all articles discussing trade policy uncertainty. *Source:* Caldara et al. 2020.

after 2017 (see Fig. 1). This increase in trade policy uncertainty itself has macroeconomic consequences. Thus, whereas much of the recent work on the effect of trade conflicts focused on the role of tariff rates (e.g., Furceri et al., 2019; Lindé and Pescatori, 2019; Jeanne and Son, 2024), we are the first to show – both empirically and theoretically – that TPU shocks are key to assessing the overall impact of trade policies on the exchange rate.

More specifically, we first estimate a structural vector autoregressive (SVAR) model of the US economy that captures TPU shocks as well as tariff shocks. Our results suggest that, in response to a TPU shock, the USD appreciates markedly against a broad currency basket. We investigate the appreciation of the USD at the height of the trade conflict and find that around 5 percentage points of the roughly 7-percent appreciation between early 2018 and 2019 can be explained by TPU shocks. In contrast, we find that shocks to the level of tariff rates have a rather small impact on the USD exchange rate.[2]

A highly conceivable explanation for the link between trade policy uncertainty and the dollar appreciation lies in safe-haven flows in response to higher uncertainty. We find evidence that deviations from the covered rate interest rate parities between US and non-US 5-year government bonds widen in response to TPU shocks and that foreign holdings of long-term US assets increase. Within the US, a rising term premium corroborates the fact that TPU matters for financial markets.

We rationalize the SVAR evidence in a two-country New Keynesian model with financial frictions and trace out the channels through which trade policy uncertainty can affect the nominal exchange rate. Crucially, the model captures an asymmetry in the relationship of the US and other countries: the notion that US dollar-denominated assets are considered to be relatively safe (see, e.g., Jiang et al., 2021; He et al., 2019; Jiang et al., 2023). We show that the financial asymmetry is a key driver of a sizable appreciation of the US dollar in the model. Simulating a shock to the uncertainty of future tariff rates, which corresponds to the observed hike in TPU in 2018–19, generates a substantial USD appreciation. Moreover, simulating a tariff rate shock that roughly corresponds to the 2018–19 increase in tariffs on aggregate US imports, our analysis suggests that TPU is a more powerful driver of USD movements than actual tariff rate hikes.

In the model, banks à la Gertler and Karadi (2011) in both countries hold domestic and foreign bonds. An agency problem between depositors and banks gives rise to a financial constraint that limits the size of the banks' balance sheet. As non-US bonds in the model are less pledgeable than US bonds, the financial constraint binds more tightly with regard to holdings of non-US bonds. This motivates a safety premium on US assets, which constitutes a deviation from the uncovered interest rate parity condition. In this setup, positive TPU shocks trigger a portfolio shift toward relatively safe US assets, raise the safety premium, and thereby cause the USD to appreciate.[3] In the model, non-financial channels also affect the transmission of uncertainty shocks, such as, for instance, a precautionary markup motive by firms. In both countries, price dynamics together with policy rate responses result in higher real short-term deposit rates. The increase in banks's funding cost eats into their net worth and forces them to sell assets to meet their financial constraint. As the constraint binds more for non-US assets, the deleveraging of banks in both countries supports the portfolio shift toward US assets.

We show that our theoretical and empirical results align well for the responses of a range of standard macroeconomic variables to TPU shocks. Importantly, in the SVAR we document that a TPU-induced USD appreciation lowers US competitiveness against the rest

[2] As trade policies targeted to Chinese products played a prominent role during the 2018–19 trade tension, we also estimate an SVAR model for the bilateral CNY/USD nominal exchange rate, employing daily data. This exercise allows us to employ data on tariff news. We find that bilateral tariff news shocks raise the USD. This broadly aligns with results in Jeanne and Son (2024). The bilateral import tariff news shocks play a more important role than aggregate import tariff rate shocks in our baseline SVAR in which we include a measure of aggregate US import tariff rates. Still, in 2018 and 2019 trade policy uncertainty shocks explain a larger fraction of the yuan/USD exchange rate than tariff news shocks according to the historical decomposition.

[3] Qualitatively similar dynamics ensue when uncertainty revolves around aggregate total factor productivity (TFP).

of the world. In particular, we find a marked decline in aggregate US import prices (roughly half the magnitude of the dollar response over the medium term) and a marked increase in aggregate foreign-currency-denominated US export prices (roughly three-quarters of the magnitude of the dollar response) in response to a positive empirical TPU shock. This translates into lower (higher) US (rest of world) consumer prices. The empirical price dynamics are present as well in our model, with the safe asset country in the model standing in for the US. There, the inflation rate differential between the two countries amplifies the appreciation of the safe asset currency. Both in the model and in the data, the US trade deficit widens in response to TPU shocks. Overall, our findings highlight a novel channel on how trade policy uncertainty counteracts potential US trade policy attempts to increase US competitiveness. This channel played an important role during the 2018–19 trade tensions.

When it comes to other model predictions, we find that the TPU-induced exchange rate change drives a divergence of output and consumption in the two countries, with consumption rising and output falling in the safe asset country. Consistent with model predictions, empirical evidence points to a contraction in global output. However, unlike in the model, US output experiences a short-lived rise in response to empirical TPU shocks while non-US output contracts over the medium term. This suggest that our model might overstate the role of the expenditure-switching channel related to exchange rate changes for quantities, as we abstract from a number of empirically relevant features to keep the model as simple as possible. This includes, for instance, stockpiling, global commodity markets, or the role of the USD as invoicing currency in trade between country pairs that do not include the US.

We show that the appreciation of the safe asset currency in response to TPU shocks in the model is robust to a number of alternative modeling choices pertaining to the presence (or absence) of capital and intermediate goods in the production function, the type of financial market integration (equity market integration or bond market integration), the pricing paradigm (dominant currency pricing or producer currency pricing), variations in the size of the safe asset currency country (accounting for half of global GDP or one sixth — closer to the case of the US), the nature of the shock as representing global trade policy uncertainty or uncertainty only with regard to the import tariffs levied by the United States.

To the extent that we investigate the effects of trade policy uncertainty, we build on Caldara et al. (2020), who construct measures of trade policy uncertainty which we employ and who analyze the role of TPU shocks to various macroeconomic variables. We expand on their work by focusing on the role of TPU for the nominal exchange rate. In their theoretical model both countries are fully symmetric, so the nominal exchange rate remains unaffected by global TPU shocks. In contrast, our model captures asymmetries between the two countries, allowing us to analyze the effect of global shocks on the exchange rate. In locating the key mechanism in the financial sector, we follow the example set by Gabaix and Maggiori (2015), who investigate the role of financial frictions for the transmission of various level shocks on the exchange rate in an open economy setting. By highlighting the effects of a second moment shock, namely the TPU shock, our analysis is complementary to their work. The role of the safe asset property of USD assets for the nominal exchange rate has previously been illustrated in a small model by Jiang et al. (2023). Our analysis differs from theirs not only by studying the transmission of a TPU shock, but also by adding more details to the model that enhance the realism of the dynamics and allow us to investigate the robustness of our main mechanism to various modeling choices as well as to generate sizable effects on the nominal exchange rate. We also complement empirical contributions that point towards a special role of the USD exchange rate and global financial markets for the transmission of US uncertainty shocks and global risk shocks (cf. Bhattarai et al. 2020, and Georgiadis et al. 2024).

The remainder of the paper is organized as follows. Section 2 presents evidence on the role of import tariffs and trade policy uncertainty for the broad USD as well as the bilateral CNY/USD exchange rate. In Section 3, we study the role of TPU shocks in an open-economy model with a safety premium on US-issued assets. The last section concludes. An online appendix contains further empirical results, robustness checks, details on the theoretical model and additional theoretical results.

2. Empirical evidence

To what extent can US trade policy explain US dollar movements? We start by tackling this question empirically and focus on the multilateral dollar exchange rate. To fully capture the effects of trade policy, in addition to the level of tariff rates, our SVAR model accounts for the uncertainty surrounding trade policy actions as a potential driver of the USD nominal exchange rate. Our results indicate that trade policy uncertainty shocks play a larger role for USD exchange rate movements than tariff rate shocks, so we focus in our exposition on trade policy uncertainty shocks and relegate the discussion of the empirical responses to tariff rate shocks to Appendix B.7. In the main exposition, we instead zero in onto the transmission of trade policy uncertainty shocks by considering the effects on a larger set of monthly macroeconomic and financial variables.

As trade policy targeted to Chinese products played a prominent role during the 2018–19 trade tensions — we also discuss the effects of trade policy uncertainty for the bilateral exchange rate vis-à-vis the Chinese yuan based on a daily data set. In this exercise, we find a more important role of – in this case bilateral – import tariff (news) shocks compared to the baseline SVAR where we include a measure of aggregate US import tariffs. Still, the effects of trade policy uncertainty dominate.

2.1. The relationship between trade policy uncertainty and the broad USD exchange rate

To examine the role of trade policy uncertainty shocks for monthly movements in the nominal effective USD exchange rate, we employ the following structural VAR framework $A_0 y_t = \alpha + A_1 y_{t-1} + \cdots + A_k y_{t-k} + \varepsilon_t$, where y_t is the vector of endogenous variables and ε_t is a vector of structural innovations with variance–covariance matrix Ξ. Identification is obtained by Cholesky factorization, i.e. by imposing a lower triangular matrix on $A = A_0^{-1}$ such that $\Sigma = AA'$, where Σ collects the reduced-form VAR residuals. The nine variables in the vector y_t are ordered as follows: (1) the US import tariff index, (2) world industrial production (excl. US), (3) US industrial production (excl. construction), (4) US consumer price index, (5) US trade policy uncertainty, (6) US macroeconomic

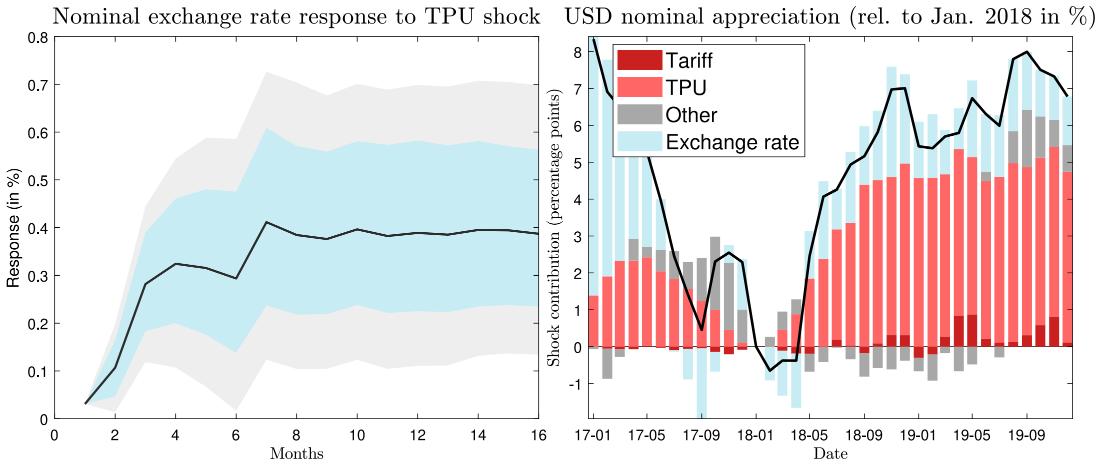

Fig. 2. Results from the monthly SVAR model. January 1985 to December 2019. Left panel: impulse response function for the US dollar appreciation given a one-standard-deviation trade policy uncertainty shock in period 1. Gray (blue) area shows the 90% (68%) confidence interval. Right panel: historical decomposition of US dollar movements for various groups of empirical shocks: tariff shocks, trade policy uncertainty (TPU) shocks, exchange rate shocks (Exchange rate), as well as other shocks and the deterministic component (Other). Percentage changes are approximated by differences of logs. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

uncertainty, (7) the US S&P 500, (8) the index of real commodity prices, and (9) the broad trade-weighted US dollar nominal exchange rate.[4] In this setup, the US import tariff index does not respond to any variable contemporaneously, meaning that tariffs rates are supposed to be set independently of the current state of the economy. Industrial production and consumer prices do not respond to the uncertainty variables within a month, implicitly assuming that real and nominal frictions in the economy hinder a quick adjustment. The financial variables (stock market prices, commodity prices, and the nominal exchange rate) are ordered last as they typically react rather swiftly to shifts in economic conditions. They are also assumed to respond to trade policy uncertainty and macroeconomic uncertainty within a month. In particular, the USD exchange rate is ordered last and responds to all variables contemporaneously.

The choice of variables follows the setup of the structural VAR employed in Bhattarai et al. (2020), who analyze the spillover of US uncertainty shocks to emerging market economies. We borrow this specification to rule out that the effect of trade policy uncertainty on the USD exchange rate is confounded with the effect of macroeconomic uncertainty. Instead of the VIX as in Bhattarai et al. (2020), we, however, include the measure of macroeconomic uncertainty estimated in Jurado et al. (2015).[5] We also include a monthly index of US import tariffs. This aims at separating direct effects of tariffs from trade policy uncertainty shocks. For trade policy uncertainty, we use the trade policy uncertainty index computed by Caldara et al. (2020) (see Fig. 1). The measure is based on a text analysis of articles in leading Anglo-American newspapers (Boston Globe, Chicago Tribune, Guardian, Los Angeles Times, New York Times, Wall Street Journal, and Washington Post). It reports the fraction of articles discussing trade policy uncertainty.[6] The model is estimated for the period of time from January 1985 to December 2019, and the lag number is set to $k = 6$. Overall, the estimated model shares similarities with the model estimated in Caldara et al. (2019).[7]

Fig. 2 plots our main empirical findings. The left panel shows the impulse response function for the US dollar effective exchange rate given a one-standard-deviation shock to trade policy uncertainty. It indicates a positive and significant effect that is rather persistent. The response of the USD exchange rate to a TPU shock is, at first glance, surprisingly sluggish. This finding is, however,

[4] All data sources are reported in Appendix A. All variables except for the macroeconomic uncertainty measure are expressed in logs. Moreover, all variables enter in first differences except the real commodity price and the nominal exchange rate. The estimation employs the BEAR toolbox (Dieppe et al., 2018). Note that the real commodity prices and the nominal exchange rate are stationary variables; transforming the data to first differences is therefore not necessary. Still, if these variables are transformed to first differences as well, the impulse response analysis yields a very similar result. Following earlier contributions, the monthly baseline VAR does not include the interest rate differential between US and non-US bonds. In Section 2.2 and Appendix B, we augment the baseline VAR by financial variables. Moreover, in analyzing the link between trade policy uncertainty and the yuan/dollar exchange rate, we rely on a (bilateral) SVAR using daily data including interest rate spreads (see Section 2.3).

[5] We include more variables in the SVAR than Bhattarai et al. (2020), in particular the US stock market index and an index of US import tariffs. In Bhattarai et al. (2020), the VIX is ordered last, whereas we order the uncertainty measures before the financial market variables. Appendix B.1 shows robustness analysis using alternative orderings of the variables.

[6] There are pros and cons to using this measure to capture trade policy uncertainty (see Deutsche Bundesbank 2020 for a discussion). One appealing advantage of the measure is that it is, by construction, exogenous with respect to macroeconomic developments. Moreover, it corresponds not solely to uncertainty in tariffs but also to uncertainty in a broader array of trade policy actions (such as import bans).

[7] Caldara et al. (2019) – utilizing the dataset introduced in Caldara et al. (2020) – also report a positive response of the nominal exchange rate to a trade policy uncertainty shock. Nevertheless, the focus of their note is neither on exchange rates dynamics, in particular, nor on offsetting effects from trade policy-induced exchange rate shifts.

in line with previous studies reporting slow responses of nominal exchange rates to fundamental shocks.[8] In part, the persistent response is due to the persistence of TPU itself to the shock, which carries over to the exchange rate. As a response to the shock, TPU hikes markedly on impact but declines quickly to a lower but still positive level, where it stays over the estimated horizon (see Appendix B.7 which shows the full set of impulse response functions).

The right panel of Fig. 2 plots the historical decomposition for the period January 2017 to December 2019. This exercise reports for each shock in the SVAR, the contribution to the percentage change of the nominal exchange rate in a specific month relative to January 2018. Between January 2018 and the end of 2018, the USD appreciated (compared to a basket of currencies) by around 6% and by one more percentage point up to the end of 2019. Importantly, according to our estimation, TPU shocks account for a large fraction of these dynamics. They explain around five percentage points of the overall change in the exchange rate up to mid-2019. Taking the direct effects of the tariff increases into account as well, US trade policy accounts for an even slightly larger fraction of the total change.

A potential concern in the specification of the benchmark SVAR model is that trade policy uncertainty might be confounded with economic policy uncertainty more generally. Conceptually speaking, trade policy uncertainty is a subset of economic policy uncertainty (EPU). In the data, the two measures could be closely related. However, when we include the Baker et al. (2016) measure of economic policy uncertainty in our SVAR model (instead of macroeconomic uncertainty), the results are qualitatively similar. The fraction of USD movements explained by TPU shocks in 2018 and 2019 is, however, slightly smaller, which likely owes to the fact that the two measures are not 100% orthogonal to each other. At the same time, shocks to economic policy uncertainty cannot explain the observed USD appreciation (see Appendix B.1). As an alternative, we conduct an even more conservative robustness analysis by orthogonalizing trade policy uncertainty with regard to economic policy uncertainty. In particular, we regress trade policy uncertainty on economic policy uncertainty (both in logs) and use the residuals from this regression as a cleaner measure of trade policy uncertainty. As reported in Appendix B.1, the response of the USD to TPU shocks is of a similar magnitude. Overall, these exercises highlight the special role of trade policy uncertainty for the USD exchange rate.

Our main result is robust to adjustments of the SVAR specification along several further dimensions (see Appendix B.1 for more details): (1) The exclusion of the Chinese yuan exchange rate from the currency basket. This robustness analysis demonstrates that the trade policy-induced multilateral appreciation in 2018 and 2019 was also the consequence of a depreciation of non-Chinese trading partners' currencies against the US dollar. (2) An estimation of the model up to the end of 2015, excluding the most recent trade policy uncertainty hikes. This exercise indicates that the role of trade policy uncertainty for the nominal exchange rate is not a peculiar phenomenon of the most recent trade conflict. (3) The inclusion of US short-term interest rate measures in the SVAR. (4) Alternative orderings of the variables included in the SVAR specification. Moreover, (5), in Appendix B.2 we account for the role of US monetary policy more explicitly by including identified monetary policy shocks from Bu et al. (2021) in a Taylor-type rule of the US shadow short rate.

Trade policy uncertainty shocks played a role for the USD exchange rate not only in 2018–19, but also in other years. However, their relevance has grown in more recent periods. This is indicated by the historical shock decomposition for the entire sample in Fig. 3. The USD appreciation is expressed in year-on-year terms, which allows for an easier interpretation of the estimates. TPU mattered, for instance, around the end of 2016 and beginning of 2017. At that time, the then newly elected US administration signaled the possibility of a tighter trade policy stance. In early 1990, TPU shocks related to US debates concerning the North American Free Trade Agreement (NAFTA) raised the value of the USD. However, other shocks are generally more important in explaining the USD exchange rate. In addition to exchange rate shocks, this includes macroeconomic uncertainty shocks and commodity price shocks. This becomes apparent, for instance, in 2008 during the global financial crisis. Shocks more closely related to fundamental driving forces of business cycles such as shocks to non-US industrial production, US industrial production and US consumer prices contribute little. This broadly aligns with insights from the literature on the exchange rate disconnect, which argues that shocks that shape business cycle dynamics only explain nominal exchange rates dynamics to a limited extent (cf. for instance Itskhoki and Mukhin 2021).

2.2. Inspecting the transmission of trade policy uncertainty shocks

To study the transmission of trade policy uncertainty shocks in more detail, the upper panel of Fig. 4 shows the response of selected variables in the baseline SVAR model to trade policy uncertainty shocks (we show the full set of responses in Appendix B.7; there we also discuss responses to tariff rate shocks and USD exchange rate shocks). In addition, in the lower panel of Fig. 4 we estimate the responses of additional variables (not included in the baseline SVAR) to trade policy uncertainty shocks. Each figure in this panel represents a separately estimated SVAR model in which we augment the baseline SVAR model by the respective variable (which we order last). The additional variables in the augmented SVARs enter in first differences of logs, except in the case of the (interest rate) spreads which enter in first differences, the nominal exchange rates which enter in logs and the real trade balance which is normalized by real GDP. The data sources are reported in Appendix A. Some of the augmented SVAR specifications have shorter samples compared to the baseline model due to data availability.

[8] For the case of monetary policy shocks, see, for instance, Müller et al. (2024) and references therein.

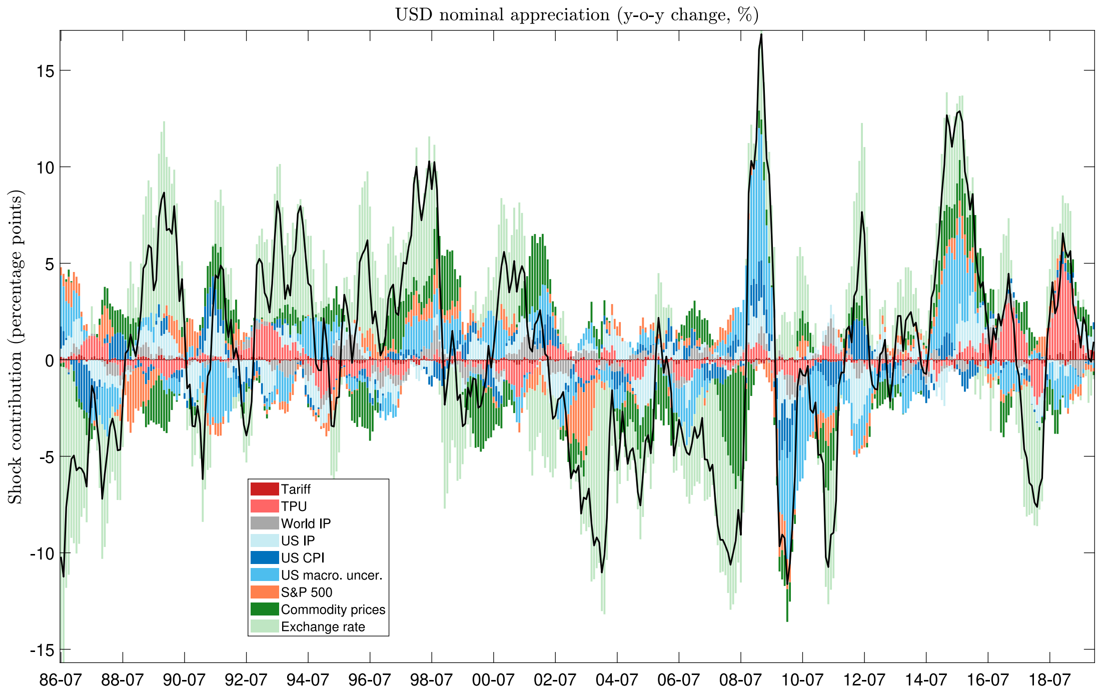

Fig. 3. Historical decomposition of US dollar movements for all shocks of the baseline monthly SVAR model. Percentage changes are approximated by differences of logs.

Macroeconomic responses

According to Fig. 4, a trade policy uncertainty shock that leads to a USD appreciation results in a decline in world industrial production and global commodity prices. Against the background of lower commodity prices and a stronger dollar, US consumer prices are also dampened.

The exchange rate channel plays an import role in the transmission of trade policy uncertainty shocks. US import prices (in USD) decline whereas US export prices expressed in foreign currency (by multiplying the USD-denominated export price index by the effective nominal USD exchange rate) rise. Rest-of-the-world CPI increases as well due to higher cost for imports from the United States (or more generally for imports denominated in US dollars) and despite the decline in global commodity prices.

Interestingly, US output does not decline in response to trade policy uncertainty shocks. Fig. 4 shows the response of a measure of monthly GDP. Industrial production shows a similar pattern; see Appendix B.7. This is likely driven by falling import prices, which support consumption and lower the cost of production. It could also be the outcome of firm and household inventory build-up. It is, however, at odds with the standard expenditure-switching mechanism of exchange rate changes, which would predict a declining demand for US goods and thus lower output given a USD appreciation. Nonetheless, the negative response of the US real trade balance (relative to GDP) after some months points to the expenditure-switching channel playing a certain role. Whereas – in line with the non-negative monthly GDP response – US consumption increases in response to positive TPU shocks, US capital good orders (as a proxy for investment) decline. This aligns with results by Caldara et al. (2020) and points toward wait-and-see behavior.

Financial market responses

In our theoretical model, the underlying mechanism translating unexpected trade policy uncertainty to movements in the USD exchange rate operates via adjustments in the global safety premium on US assets. To shed light on the response of the safety premium to empirical TPU shocks, we augment our SVAR with a measure for deviations from the covered interest parity (CIP) between non-US and US government bonds. For this, we aggregate country-specific CIP deviations provided by Du et al. (2018) and Du and Schreger (2016) using USD exchange rate basket weights. The aggregate CIP deviation serves as a proxy for changes in the safety premium.[9]

Fig. 4 shows that the aggregate CIP deviation widens in response to a TPU shock. The response is – at 68% confidence bands – statistically significant. The positive reaction is quite persistent, mirroring the similarly sluggish pattern of the US dollar response. According to the historical decomposition (see Appendix B.4), in 2018 the TPU shock adds up to above 5 basis points to the CIP

[9] See Appendix B.4 for more details. As we are mainly interested in the implications of TPU shocks for the broad USD, we exclude observations on Chinese government bonds. In this SVAR specification (and in the exercise in which we study the response of changes in foreign long-term US asset holdings), we additionally control for five-year Treasury yields.

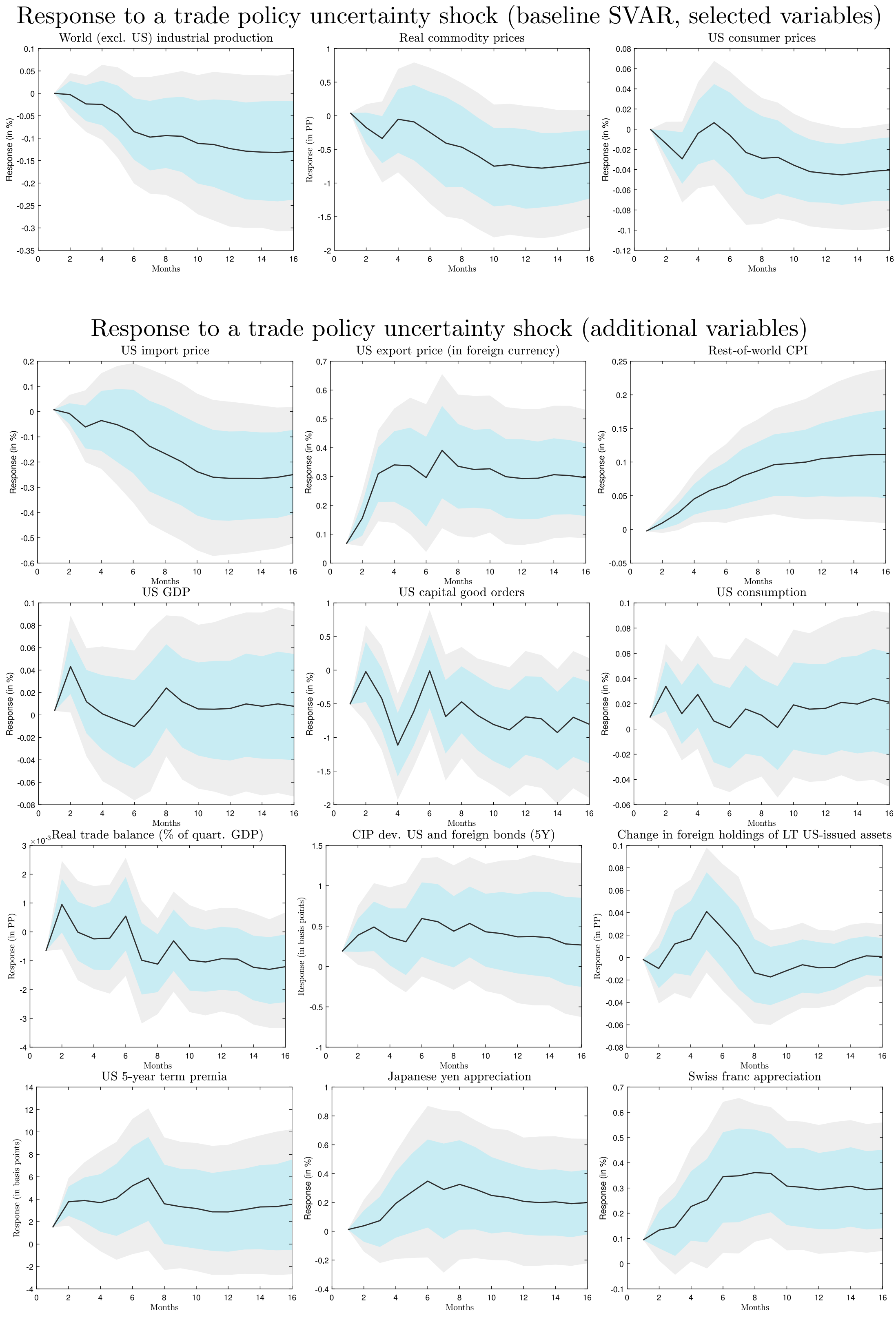

Fig. 4. Monthly SVAR (January 1984 to December 2019). Some specifications have shorter samples due to data limitations; see text for more details. IRF of selected variables given a one-standard-deviation trade policy uncertainty shock in period 1. Gray (blue) area shows the 90% (68%) confidence interval. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

deviation between non-US and US government bonds; at the end of 2019, the effect is still around 2 1/2 basis points. In an additional exercise in Appendix B.4, we also utilize the panel structure of the data set of Du et al. (2018) and Du and Schreger (2016). Using local projections, we report that, in response to trade policy uncertainty shocks, the US Treasury premium increases for long-term bonds of advanced and emerging economies alike. Overall, the findings indicate that CIP deviations between long-term non-US and US government bonds play a relevant role in the transmission of TPU shocks.

In the US itself, trade policy uncertainty shocks raise the term spread expressed by the spread between five-year Treasuries and the Federal Funds Rate. Moreover, data on US cross-border financial transactions suggests an increase in the relative demand for US-issued long-term assets. In particular, Fig. 4 shows a positive change of foreign long-term US asset holdings — measured in real terms and as three-month moving averages to smoothen the very volatile series. In Appendix B.5, we use local projections to study the TPU-shock-induced effects on US banking variables available at quarterly frequency, in particular net worth and asset holdings, as well as the leverage ratio.

To compare the USD response to a TPU shock with the response of other currencies considered to be relatively safe, we separately augment the baseline SVAR model with the nominal effective exchange rates of the Swiss franc and the Japanese yen. In both cases, we include a broad exchange rate index excluding the USD (while we leave the USD broad nominal exchange rate in the specification). Fig. 4 indicates that both currencies also appreciate multilaterally in response to a TPU shock. This suggests that the TPU shocks also trigger relative demand shifts towards assets issued in other safe-haven currencies. Repeating the same exercise with the effective exchange rates of the euro and pound sterling indicates a different pattern. Both the euro and the pound tend to depreciate after TPU shocks (see Appendix B.6.2). This is reasonable as their relevance as safe haven currencies is more limited (cf. Georgiadis et al. 2024, who find that the euro and the pound depreciate after global risk shocks).[10]

Our results thus suggest that particularly issuers of global reserve currencies such as the US are prone to the phenomenon of a trade-policy-uncertainty appreciation of their own currencies.

2.3. The Chinese yuan / US dollar bilateral exchange rate response to trade policy uncertainty shocks

As a final empirical exercise, we use daily data on trade policy measures and financial market variables to estimate a structural VAR that includes the bilateral CNY/USD exchange rate.[11] Compared to the baseline monthly SVAR model in which we use a measure of actual aggregate tariff rates for the US, this exercise has the advantage that actual tariff shocks are larger and that we have information on tariff-related announcements on a daily basis. It thus serves as a robustness check to assess the role played by trade policy uncertainty shocks in comparison to actual tariff shocks.

We again use a Cholesky factorization as an identifying restriction. The vector of variables includes, in the following order: (1) a measure quantifying announcements and news regarding US tariffs, specifically on Chinese imports in 2018–20 (see Appendix A.2 for details), (2–3) the Citigroup macroeconomic surprise indices for the US and China, (4) trade policy uncertainty (reported at a daily frequency from Caldara et al. 2020), (5) the CBOE Volatility Index VIX, (6) the difference between the US and Chinese short-term policy rates, (7) the spread between the yields on US and Chinese government bonds with a three-year maturity, (8) the difference in the growth rates of stock market values in the two countries measured by the S&P 500 for the US and the Shanghai SE Composite Index for China as well as (9) nominal oil prices (Brent) and (10) the nominal exchange rate between the USD and the renminbi.[12] As in the monthly VAR, we allow each variable to contemporaneously react to all variables ordered subsequently. In particular, the CNY/USD exchange rate responds to all variables within a day. The model is estimated for the period of January 1, 2014 to March 5, 2020. We exclude observations in which any data are missing (for instance on weekends). Given the choice of high-frequency financial variables and news indicators, we set the lag number to two working weeks ($k = 10$). All data sources are reported in Appendix A.

Fig. 5 plots the results. We find a positive and significant response of the CNY/USD nominal exchange rate to US TPU shocks.[13] Even though a very different set of variables is employed in the SVAR and we analyze the bilateral CNY/USD exchange rate instead of the multilateral USD exchange rate, the historical decomposition in the right panel suggests a very similar conclusion to the broader analysis above. In particular, between 2018 and 2019, a large fraction of the USD appreciation against the renminbi seems to have been triggered by trade policy uncertainty and tariff news shocks. In this specification, tariff news shocks play a more substantial role than the tariff shocks in the monthly model, presumably reflecting the fact that most trade policy measures taken in this period were directed against China.[14] Nonetheless, the effect of TPU shocks remains dominant. Adding up the contributions

[10] In Appendix B.6.3, we study the role of TPU and UK-specific economic policy uncertainty (that is in many episodes correlated to Brexit-related uncertainty) for the pound sterling. We find that – different than for the US dollar – (i) the pound depreciates in response to trade policy uncertainty shocks and in response to UK-specific economic policy uncertainty shocks and (ii) both shocks explain to some extent the large depreciation observed since mid-2016.

[11] Employing the CNY/USD exchange rate in a VAR with monthly data on a longer sample faces several challenges, as the exchange rate was inflexible throughout many episodes. Bearing this caveat in mind, Appendix B.6.1 reports results from a monthly SVAR specification including the bilateral yuan/USD exchange rate and estimated on a shorter sample compared to the baseline monthly model.

[12] The interest rate spreads (which appear to have downward trends in the sample) as well as the stock market variables are transformed to first differences. All variables except the tariff news measure and the surprise indices enter in logs.

[13] The magnitudes of the estimates in the left panels of Figs. 2 and 5 are not comparable given the different frequency of the data.

[14] Despite the very limited time series variation of this measure, the effect of a tariff news shock on the nominal exchange rate is – based on 68% error bands – statistically significant over the whole horizon. Within 90% error bands, the effect is still statistically significant for two months. See Appendix B.3.1.

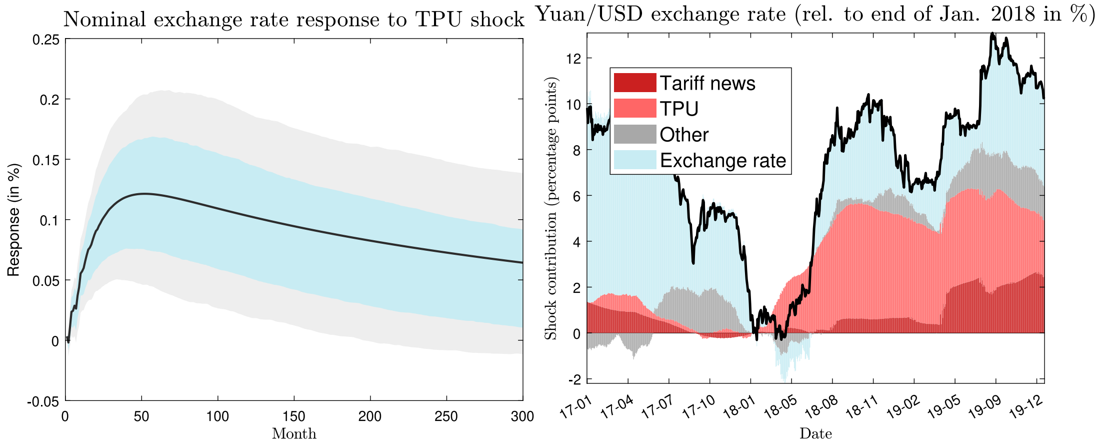

Fig. 5. Results from the daily SVAR model. January 2014 to March 2020. Left panel: impulse response function for CNY/USD nominal exchange rate appreciation given a one-standard-deviation trade policy uncertainty shock in period 1. Gray (blue) area shows the 90% (68%) confidence interval. Right panel: historical decomposition of US dollar movements for various groups of empirical shocks: tariff shocks, trade policy uncertainty shocks (TPU), exchange rate shocks (Exchange rate), as well as remaining shocks and the deterministic component (Other). Percentage changes are approximated by differences of logs. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

of both types of trade-related shocks, we arrive at the conclusion that, between end-of-January 2018 and the end of 2019, the USD appreciated against the renminbi by around 5 to 6% due to US trade policy.[15]

3. A model of trade policy (uncertainty) and the dollar exchange rate

As our empirical result is novel, this section proposes a theoretical model that rationalizes the link between an increase in trade policy uncertainty (TPU) and the appreciation of the US dollar. The two-country DSGE model features financial frictions à la (Gertler and Karadi, 2011) and international trade in government bonds. The key mechanism is based on the notion that assets issued in US dollars are relatively safe by international comparison (see, e.g., Jiang et al., 2021; He et al., 2019; Jiang et al., 2023). In the model, this implies that risk-averse investors shift their asset demand toward US dollar-denominated assets in the face of an increase in uncertainty. This portfolio shift supports an appreciation of the US dollar. We show that the link between TPU and USD appreciation breaks down when the safe asset channel is removed from the model. In contrast, when the channel is active, the link survives various alternative modeling choices. The effect of tariff rate shocks on the exchange rate is muted and barely altered by the presence of financially constrained banks.

3.1. The model

Both countries' economies feature households, firms, banks, a central bank, and a fiscal authority. The key asymmetry between the countries is the difference in the safety of assets. Secondly, motivated by the dominance of the US dollar in international trade, we assume that US exporters commit to producer currency pricing (PCP), while exporters from the second country follow local currency pricing (LCP).[16] As many model features are standard, this section provides only a brief summary and highlights the key equations. We largely focus on discussing the economy of country A, which stands for the US in the model, marking the variables with the superscript A. Features of country B are only discussed in those cases in which they deviate from their country A counterpart. The full model and derivations are relegated to Appendix C.

Households. Households consume, supply labor to firms and save in domestic deposits. Their utility function is standard and separable in consumption and leisure. The household's budget constraint reads $C^A_{i,t} + D^A_{i,t} = R^A_{d,t-1} D^A_{i,t-1} + \frac{W^A_{i,t}}{P^A_t} L^A_{i,t} + T^A_{i,t}$, where $R^A_{d,t}$ is the real interest rate on the household's deposits with the domestic bank, $D^A_{i,t}$. $W^A_{i,t}$ is the nominal wage set by household i, P^A_t is the consumer price index, and $T^A_{i,t}$ is a term summarizing the net income from transfers, taxes, and firms' profits. In equilibrium, all households face the same first-order conditions for consumption and saving. Each household supplies labor equally to a continuum of unions j, $j \in [0,1]$. Monopolistically competitive unions offer differentiated labor services, $L^A_{j,t}$, to firms at wage $W^A_{j,t}$. The aggregation of labor services follows a Dixit–Stiglitz aggregator and wage setting is subject to nominal rigidities à la Calvo (1983) such that a standard wage Philips curve arises.

[15] It is often argued that the CNY/USD exchange rate is, at least to a certain extent, possibly not market-driven as Chinese authorities might regularly intervene in the exchange rate market. Such exogenous interventions would be captured by the empirical exchange rate shocks (blue bar in Fig. 5, right panel). However, these exchange rate shocks are not necessarily exchange rate interventions but can be related to many other determinants of the CNY/USD exchange rate.

[16] For a discussion of the dominance of the US dollar in trade, see Gopinath et al. (2020).

Firms. In the baseline version of the model, firms in each economy employ only labor as input for the production of differentiated final goods.[17] The production function of firm j reads $Y^A_{p,jt} = A^A_t L^A_{j,t}$. Here, $Y^A_{p,jt}$ denotes the production volume and A^A_t denotes the total factor productivity. Firms produce for the domestic as well as for the export market. Their profit function reads $P^{A,A}_{j,t} Y^{A,A}_{j,t} + \mathcal{E}^{A,B}_t P^{A,B}_{j,t} Y^{A,B}_{j,t} - W^A_{t+k} L^A_{j,t}$. $P^{A,A}_{j,t}$ is the price that firm j sets for its goods sold at home. $P^{A,B}_{j,t}$ is the pre-tariff price for its goods in the export market, in terms of the currency of country B. $\mathcal{E}^{A,B}_t$ is the bilateral nominal exchange rate between country A and country B.

Firms are in monopolistic competition and set their prices with a markup over their marginal costs. They face price rigidities à la (Calvo, 1983) with the probability of not being able to reset their price being ζ. We assume that firms in country A set their optimal export prices in their own currency, according to the producer currency price paradigm. Motivated by the dominance of the US dollar, the currency of country A, in international goods trade, we assume that firms in country B set their export prices in terms of the currency of their export market, following the local currency paradigm.

Aggregate demand and price indices. In the baseline model, aggregate demand on the goods market, Y^A_t, is composed of consumption and government spending, G^A_t. Thus, for country A, it holds that $Y^A_t = C^A_t + G^A_t$. The aggregate demand aggregator in country A reads $Y^A_t = \left[(1-\mu^T)^{1/\Theta}(Y^{AA}_t)^{\frac{\Theta-1}{\Theta}} + (\mu^T)^{1/\Theta}(Y^{BA}_t)^{\frac{\Theta-1}{\Theta}}\right]^{\frac{\Theta}{\Theta-1}}$. Y^{AA}_t denotes the goods produced in A and absorbed in A, while Y^{BA}_t denotes goods produced in country B and imported by A. Parameter μ^T governs the trade openness, and Θ is the elasticity of substitution between goods produced in country A and goods produced in country B. The aggregate price index in country A, P^A_t, is $P^A_t = \left[(1-\mu^T)(P^{AA}_t)^{1-\Theta} + \mu^T(\widetilde{P}^{BA}_t)^{1-\Theta}\right]^{\frac{1}{1-\Theta}}$, where P^{AA}_t is the price level for goods produced in country A in the domestic market and $\widetilde{P}^{BA}_t$ is the post-tariff price for goods produced in country B and used in country A, in terms of the currency of country A. It holds that $\widetilde{P}^{BA}_t = (1+\tau^{BA}_t)P^{BA}_t$, where τ^{BA}_t is the import tariff rate charged on goods from country B by the customs authority of country A, and P^{BA}_t is the respective pre-tariff price.[18]

Financial sector. The banking sector is modeled in the vein of Gertler and Karadi (2011, 2013). Banks in both countries fund themselves with deposits from domestic households and invest in domestic as well as foreign assets. In the baseline version of the model, the internationally traded assets are long-term government bonds. An alternative version in Section 3.3.4 considers trade in claims on capital (equity). The choice of asset type does not affect the structure of the banks' optimization problem. The balance sheet of bank j reads

$$Q^A_{b,t} B^{AA}_{jt} + \widetilde{Q}^B_{b,t} B^{BA}_{jt} = N^A_{jt} + D^A_{jt}.$$

Here, $B^{AA}_{j,t}$ denotes assets issued in country A and held by banks in country A. $B^{BA}_{j,t}$ are assets issued in country B and held by banks in country A. $Q^A_{b,t}$ is the real price of bonds issued in country A. $\widetilde{Q}^B_{b,t} = RER_t Q^B_{b,t}$ is the real price of bonds issued in country B in terms of goods in country A. $N^A_{j,t}$ is the bank's net worth and D^A_{jt} is its deposits. Banks add the profits that they earn on their assets to their net worth, which evolves according to

$$N^A_{jt} = (R^A_{bt} - R^A_{d,t-1})Q^A_{t-1} B^{AA}_{j,t-1} + (\widetilde{R}^B_{bt} - R^A_{d,t-1})\widetilde{Q}^B_{t-1} B^{BA}_{j,t-1} + R^A_{d,t-1} N^A_{j,t-1}.$$

R^A_{bt} is the real return on bonds issued in country A, R^A_{dt} is the return of deposits at banks in country A, and $\widetilde{R}^B_{bt} = \frac{RER_t}{RER_{t-1}} R^B_{bt}$ is the real return on bonds issued in country B in terms of the currency of country A. Following Woodford (1998, 2001), we model long-term government bonds as geometrically decaying consols with a fixed coupon, c_b, and a decay rate, ρ_b. The average duration of such a bond is $1/(1-\beta\rho_b)$. Accordingly, the real return on long-term government bonds issued in country A to country-A banks is

$$R^A_{bt} = \frac{c_b + \rho_b Q^A_t}{Q^A_{t-1}}. \tag{1}$$

Banks accumulate wealth until they exit the financial sector. Their exit probability is exogenously given by θ and ensures that bankers do not reach the point where they outsave their financial constraint. Bankers choose their asset holdings such as to maximize the expected terminal wealth at the point of exit,

$$V^A_{jt} = \beta E_t \Lambda^A_{t+1}[(1-\theta)N^A_{j,t+1} + \theta V^A_{j,t+1}]. \tag{2}$$

The terminal net worth of bankers that exit the sector is distributed across households as lump-sum income. In turn, households provide bankers that newly enter the business with initial capital.

[17] We show that the main result holds in extensions of this model in which firms have access to capital (see Section 3.3.4) or intermediate goods (see Section C.5.1) as inputs in their production function.

[18] Our setup implies a complete pass-through of tariff rate changes to prices of final goods. While this might only apply to selected cases of US imports, the results by Caldara et al. (2020) show the importance of TPU also with an incomplete tariff rate pass-through. In their model, tariff rate increases affect the costs of wholesale firms, which purchase domestic and foreign intermediate goods. These firms sell their output domestically setting their prices subject to adjustment costs. In that case, price stickiness reduces the tariff rate pass-through. In contrast, in our model, price setting takes place on the exporters' side. This yields the advantage that we are able to investigate alternative optimal reactions to exchange rate movements by forward-looking price setters (local currency pricing vs. producer currency pricing).

The contract between bankers and depositors is subject to a costly enforcement/moral hazard problem. Each period, the banker has the option to divert the assets for private purposes. If a banker does so, she is forced into insolvency by the depositors, who can only recover a fraction of each asset type, depending on its pledgeability. To ensure that bankers abstain from diverting assets, the value of staying in business must always be greater than or equal to the fraction of assets that the banker can divert.[19] The incentive compatibility constraint reads

$$V_{jt}^{A} \geq \lambda_{AA} Q_{b,t}^{A} B_{jt}^{AA} + \lambda_{BA} \widetilde{Q}_{b,t}^{B} B_{jt}^{BA}. \tag{3}$$

Parameters λ_{AA} and λ_{BA} govern the pledgeability of the respective assets for banks in country A. Following the literature on safe assets in international finance (see, e.g., Jiang et al., 2021; He et al., 2019; Jiang et al., 2023), we assume that dollar assets, i.e. assets issued in country A are relatively safe and more pledgeable than assets issued in country B, i.e. $\lambda_{AA} < \lambda_{BA}$.[20]

To solve the bankers' optimization problem, we guess that the value function is linear in holdings of government bonds of both countries as well as net worth

$$V_{jt}^{A} = v_{bjt}^{AA} Q_{t}^{A} B_{jt}^{AA} + v_{bjt}^{BA} \widetilde{Q}_{t}^{B} B_{jt}^{BA} + v_{njt}^{A} N_{jt}^{A}. \tag{4}$$

In optimum, the coefficients v_{bjt}^{AA}, nu_{bjt}^{BA}, and v_{njt}^{A}, are shadow values for the bank of an additional unit of domestic bonds, foreign bonds or net worth, respectively. In this setting, the first-order conditions for holdings of domestic and foreign bonds as well as for the Lagrangian multiplier on the incentive constraint, μ_{jt}^{A}, can be obtained as

$$v_{bjt}^{AA} = \lambda_{AA} \frac{\mu_{jt}^{A}}{1+\mu_{jt}^{A}}, \tag{5}$$

$$v_{bjt}^{BA} = \lambda_{BA} \frac{\mu_{jt}^{A}}{1+\mu_{jt}^{A}}, \tag{6}$$

$$Q_{t}^{A} B_{jt}^{AA} = \frac{v_{bjt}^{BA} - \lambda_{BA}}{\lambda_{AA} - v_{bjt}^{AA}} \widetilde{Q}_{t}^{B} B_{jt}^{BA} + \frac{v_{njt}^{A}}{\lambda_{AA} - v_{bjt}^{AA}} N_{jt}^{A}. \tag{7}$$

Verifying the guess for the value function results in

$$v_{bjt}^{AA} = \beta E_t \Omega_{j,t+1}^{A} (R_{b,t+1}^{A} - R_{d,t}^{A}), \tag{8}$$

$$v_{bjt}^{BA} = \beta E_t \Omega_{j,t+1}^{A} (\widetilde{R}_{b,t+1}^{B} - R_{d,t}^{A}), \tag{9}$$

$$v_{njt}^{A} = \beta E_t \Omega_{j,t+1}^{A} R_{d,t}^{A}, \tag{10}$$

where we define

$$\Omega_{j,t}^{A} \equiv \Lambda_{t}^{A}((1-\theta) + \theta(1+\mu_{jt}^{A}) v_{njt}^{A}). \tag{11}$$

For aggregation, we assume an equilibrium in which all banks are symmetric (i.e., $\forall j : v_{bjt}^{AA} = v_{bt}^{AA}, v_{bjt}^{BA} = v_{bt}^{BA}, v_{njt}^{A} = v_{nt}^{A}, \Omega_{jt}^{A} = \Omega_{t}^{A}$). The optimization problem and its solution for the banks in country B take an analogous form. The full derivation of the banker's solution and further equations relating to the aggregation of the banking sector are relegated to Appendix C.2.

Trade policy uncertainty, fiscal and monetary authority. The key fiscal instruments in this model are import tariffs. The import tariff rate on goods produced in country B and imported by country A, τ_t^{BA}, and the import tariff rate on goods produced in country A and imported by country B, τ_t^{AB}, follow exogenous AR(1) processes with persistence parameter ρ_τ and $\epsilon_t^{\tau,BA}$ and $\epsilon_t^{\tau,AB}$ as tariff rate shocks.

$$\tau_t^{BA} = \rho_\tau \tau_t^{BA} + \epsilon_t^{\tau,BA}, \qquad \epsilon_t^{\tau,BA} \sim \mathcal{N}(0, \sigma_t^{\tau}) \tag{12}$$

$$\tau_t^{AB} = \rho_\tau \tau_t^{AB} + \epsilon_t^{\tau,AB}, \qquad \epsilon_t^{\tau,AB} \sim \mathcal{N}(0, \sigma_t^{\tau}). \tag{13}$$

We follow Caldara et al. (2020) and model a trade policy uncertainty shock as a shock, $\epsilon_t^{\sigma,\tau}$, that increases the standard deviation of tariff rate shocks, hence widening the path of future tariff rates,

$$\sigma_t^{\tau} = \rho_{\sigma,\tau} \sigma_{t-1}^{\tau} + \epsilon_t^{\sigma,\tau}. \tag{14}$$

Apart from import tariff rates, governments fund their exogenous government spending G_t by issuing government bonds, which are bought by banks of both countries, as well as by lump sum taxes, T_t^A, which follow a feedback rule and are sensitive to the level of public debt. In summary, $G_t^A + R_{b,t}^A Q_{t-1}^A B_{t-1}^A = Q_t^A B_t^A + T_t^A + \tau_t^{BA} P_t^{BA} Y_t^{BA}$ is the government's flow budget constraint.

The central bank sets the short-term nominal interest rate by following the Taylor-type rule $R_{n,t}^A = (R_{n,t-1}^A)^\rho \left(\left(\frac{\Pi_t^A}{\Pi^A}\right)^{\phi_\pi} \left(\frac{Y_t^A}{Y^A}\right)^{\phi_y} \right)^{(1-\rho)}$, where $R_{n,t}^A = R_{d,t}^A E_t[\Pi_{t+1}^A]$. Parameter ρ is the degree of interest rate smoothing. ϕ_π and ϕ_y govern the feedback of the policy rule to inflation and output, respectively.

[19] Throughout our experiments we assume that this relationship holds with equality.

[20] Distinguishing assets of different countries in a setting with banks à la (Gertler and Karadi, 2011) by their pledgeability, goes back to Trani (2015). In the closed economy context, Gertler and Karadi (2013) use this modeling approach to distinguish between riskier capital assets and more pledgeable government bonds. Meeks et al. (2017) specify different pledgeability parameters to distinguish the characteristics of standardized asset-backed securities and opaque loans.

Table 1
Calibration.

Symbol	Parameter	Value
σ_c	Coefficient of relative risk aversion	2
σ_l	Inverse Frisch elasticity	1
β	Discount factor	0.995
h	Habit formation	0.75
ϵ	Elasticity of goods demand	6
ϵ_w	Elasticity of labor demand	6
Θ	Elast. of substit. between A-goods and B-goods	1.5
μ^T	Trade openness	0.15
ζ	Price Calvo parameter	0.75
ζ_w	Wage Calvo parameter	0.75
ρ	Taylor rule: interest rate smoothing	0.8
ϕ_π	Taylor rule: inflation coefficient	2
ϕ_y	Taylor rule: output coefficient	0.125
G/Y	Government spending share	0.2
$B/(4Y)$	Debt-to-annual-GDP ratio	1
c_b	Coupon on government bond	0.2
ρ_b	Decay rate of government bond	0.96
κ_τ	Tax rule: coefficient on government debt	0.2
μ^A	Asset openness	0.25
LEV	Leverage ratio	10
θ	Survival rate of banker	0.95
λ_{AA}	Pledgeability parameter of asset A held by A-banks	0.15
λ_{AB}	Pledgeability parameter of asset A held by B-banks	0.148
λ_{BA}	Pledgeability parameter of asset B held by A-banks	0.20
λ_{BB}	Pledgeability parameter of asset B held by B-banks	0.197
$R_b^B - R_d$	Spread: return on B-assets over deposit rate	25 bp
$R_b^B - R_b^A$	Spread: return on B-assets over return on A-assets	6.25 bp
ρ_τ	Persistence of tariff shock	0.99
σ_τ	Standard dev. of tariff rate shock	0.01
ρ_{σ_τ}	Persistence of tariff rate uncertainty shock	0.96

International linkages. We assume that both countries are the same size. They are linked through trade in goods and assets. The real trade balance of country A reads

$$TB_t^A = Y_t^{AB}\widetilde{P}_t^{AB,B}RER_t - Y_t^{BA}\widetilde{P}_t^{BA,A}. \tag{15}$$

$\widetilde{P}_t^{AB,B}$ is the post-tariff price of goods exported from A to B relative to the price level in country B. The adjustment by the real exchange rate guarantees that the trade balance of country A is real in terms of its own currency and price level. Conversely, $\widetilde{P}_t^{BA,A}$ is the post-tariff price of goods exported from B to A relative to the aggregate price level in A.

The assumption that government bonds can be traded internationally and are held by banks in both countries implies the market clearing conditions for bonds $B_t^A = B_t^{AA} + B_t^{AB}$ and $B_t^B = B_t^{BA} + B_t^{BB}$, where B_t^A and B_t^B are bonds issued by the government in countries A and B, respectively. B_t^{AA} and B_t^{AB} are bonds that are issued in A and held by banks in countries A and B, respectively. Analogously, B_t^{BA} and B_t^{BB} are bonds issued by government B and held by banks in countries A and B, respectively. The evolution of the net foreign asset position of country A is tied to its trade balance according to

$$(\widetilde{Q}_t^B B_t^{BA} - Q_t^A B_t^{AB}) = (\widetilde{R}_{b,t}^B \widetilde{Q}_{t-1}^B B_{t-1}^{BA} - R_{b,t}^A Q_{t-1}^A B_{t-1}^{AB}) + TB_t^A. \tag{16}$$

3.2. Calibration

Key to our result is the asymmetric financial friction, which generates a steady-state convenience yield. The pledgeability parameters in the incentive constraint (Eq. (3)), λ_{AA}, λ_{BA}, and their country-B counterparts, are pinned down by the steady-state interest rates, the steady-state leverage ratio of banks, the survival probability of bankers, and their steady-state asset holdings.[21] Table 1 provides the calibration of the parameters of the baseline model. We set the steady-state leverage ratio to 10. This is below the empirically observed average for the leverage ratios of US banks.[22] However, for an asset-to-equity ratio of 15 or 20, the determinacy of equilibrium dynamics would be violated in the model. On the other hand, this value is higher than in the paper by Gertler and Karadi (2011), who fix the steady-state leverage ratio at 4. Their choice is motivated by an economy-wide average that extends to non-financial firms with a low asset-to-equity ratio. The steady-state deposit rate in both countries is pinned down by the calibration of β. The steady-state spread of the return on country-B government bonds over the deposit rate is set to 100 basis points at an annualized rate, following Gertler and Karadi (2011, 2013). The asymmetry between the pledgeability parameters in

[21] For further details, see Appendix C.2.

[22] See Board of Governors of the Federal Reserve System (2022).

the incentive constraint reflects the asymmetry in the steady-state excess returns of country-A and country-B assets over the deposit rates. For country A, the steady states of Eqs. (5), (6), (8), and (9) imply that $\frac{\lambda_{AA}}{\lambda_{BA}} = \frac{(R_b^A - R_d^A)}{(\widetilde{R}_b^B - R_d^A)}$. We set the annualized steady-state difference between the returns on A-bonds and B-bonds to 25 basis points. This reflects that investors are, on average, willing to give up 25 basis points per annum for holding safe US treasuries as discussed in Jiang et al. (2021). The survival rate of bankers is set to $\theta = 0.95$. The calibration implies that the pledgeability parameters of A-assets and B-assets for banks in country A are, respectively, roughly $\lambda_{AA} = 0.15$ and $\lambda_{BA} = 0.20$.[23] The steady-state trade deficit for country A is set to two percent of its GDP, which is roughly in line with the US trade deficit. This implies that country-A banks have to earn more on their assets than country-B banks to balance the current account (Eq. (16)) and that they hold more assets. $\mu_A = 0.25$ governs the openness for trade in assets. As a consequence, 75% of country-A assets are held by country-A banks. Banks in both countries have a substantial home bias in asset holdings: domestic assets make up 58% of the portfolio by country-A banks and 64% of the portfolio by country-B banks respectively. These values fall within the range of values reported for home bias in developed economies by Coeurdacier and Rey (2013).

In our baseline model, banks' cross-border asset trade is in long-term government bonds. Hence, the model entails a fiscal sector. The share of government spending in output is set to 20%. We assume a debt-to-GDP ratio of 100%, roughly in line with current total public debt in the US at the height of the 2018–19 trade tensions.[24] The geometric decay rate of the consol, $\rho_b = 0.96$ implies an average duration of the bond of five years. The coefficient on government debt in the tax rule, κ_τ, is set to ensure determinacy of equilibrium dynamics.

The persistence parameters for the shock processes for the tariff rate and the tariff rate uncertainty, $\rho_\tau = 0.99$ and $\rho_{\sigma_\tau} = 0.96$, are taken from Caldara et al. (2020).

The other parameters in the model are calibrated to standard values and symmetric across countries. Following Caldara et al. (2020), we set the coefficient of relative risk aversion $\sigma_c = 2$, and the inverse of the Frisch elasticity $\sigma_l = 1$. The discount factor $\beta = 0.995$ implies that the annualized steady-state real interest of deposits is 2%. $h = 0.75$ implies a persistent habit formation in consumption. Likewise, the elasticity of substitution between varieties of goods produced in one country, $\epsilon = 6$, and between varieties of labor, $\epsilon_w = 6$, fall within the range of values commonly adopted in the literature. ζ and ζ_w denote the probabilities for each firm and each union to adjust their prices or wages, respectively, in any given period. The value of 0.75 implies an average duration for prices and wages of one year. In our parametrization of the trade openness and the elasticity of substitution between goods produced in country A and goods produced in country B, we follow Caldara et al. (2020) and set $\mu^T = 0.15$ and $\Theta = 1.5$. The Taylor rules of both central banks feature substantial interest rate smoothing ($\rho = 0.8$) as usually diagnosed in the context of estimations of structural models.[25] $\phi_\pi = 2$ and $\phi_y = 0.125$ are standard values for the feedback coefficients.

3.3. Trade policy shocks and the US dollar appreciation in the model

Our theoretical analysis focuses on *synchronized*, i.e. global, trade policy shocks in both countries. Before we highlight our key results regarding the role of an increase in global trade policy uncertainty – which are second-moment shocks – we show the responses to actual first-moment shocks, namely global tariff rate increases. In all exercises, we compare the effects in two model versions with and without the asymmetric financial friction. We show that a safe asset channel in the baseline model can explain the results of our SVAR that TPU shocks can trigger an appreciation of the safe asset currency. The model also generates an appreciation of the safe asset currency after a tariff rate shock. Calibrating the shocks such that they reflect the episode of the 2018–19 trade tensions between the US and its major trading partners, we find that the effects of trade policy uncertainty shocks on the exchange rate in the model are more pronounced than the effects of tariff rate hikes. In Appendix C.6.3, we show that a unilateral tariff rate increase by the safe asset country causes a more notable appreciation of its currency. Nonetheless it falls short of the appreciation after the TPU shock. Appendix C.5 documents the robustness of the link between trade policy uncertainty and the safe asset appreciation to various alternative modeling choices.

3.3.1. The effect of a synchronized tariff rate increase

Our empirical results imply that realized tariff rate shocks contributed rather little to the multilateral appreciation of the US dollar at the height of the trade conflict with China. In that trade conflict, China retaliated to new US tariffs by raising its own import tariffs on US goods. Accordingly, this section analyzes the effect of a shock that raises import tariff rates in both countries.[26] The size of the shock is set to 2 percentage points. This roughly accounts for the 2018–19 increase in the import tariff rate for *aggregate* US imports.

Fig. 6 shows impulse responses to the symmetric first-moment tariff rate shocks, both in the full model (red lines) and in a model without financial frictions (blue lines). Absent the asymmetric financial friction, the effects in both countries are symmetric. The shock triggers an increase in consumer prices in both countries. Consequently, both central banks react by raising their respective policy rate. Consumption and output fall in both countries. Since the story is exactly the same in both countries, nominal interest

[23] As the portfolios of banks in both countries differ due to their home bias, the values of the pledgeability parameters for both assets in country B differ slightly, with $\lambda_{AB} = 0.148$ (for assets issued in country A and held by banks in country B) and $\lambda_{BB} = 0.197$.

[24] See https://fred.stlouisfed.org/series/GFDEGDQ188S.

[25] See, e.g., Smets and Wouters (2007) or, in more recent estimations on US data, Kulish et al. (2017) or Boehl and Strobel (2024).

[26] Here, we analyze the effect of first-order tariff rate shocks in a log-linear setting. Appendix C.6.3 simulates a tariff rate shock with a third-order approximation. The difference of the third-order IRFs from the first-order IRFs presented here is, however, negligible.

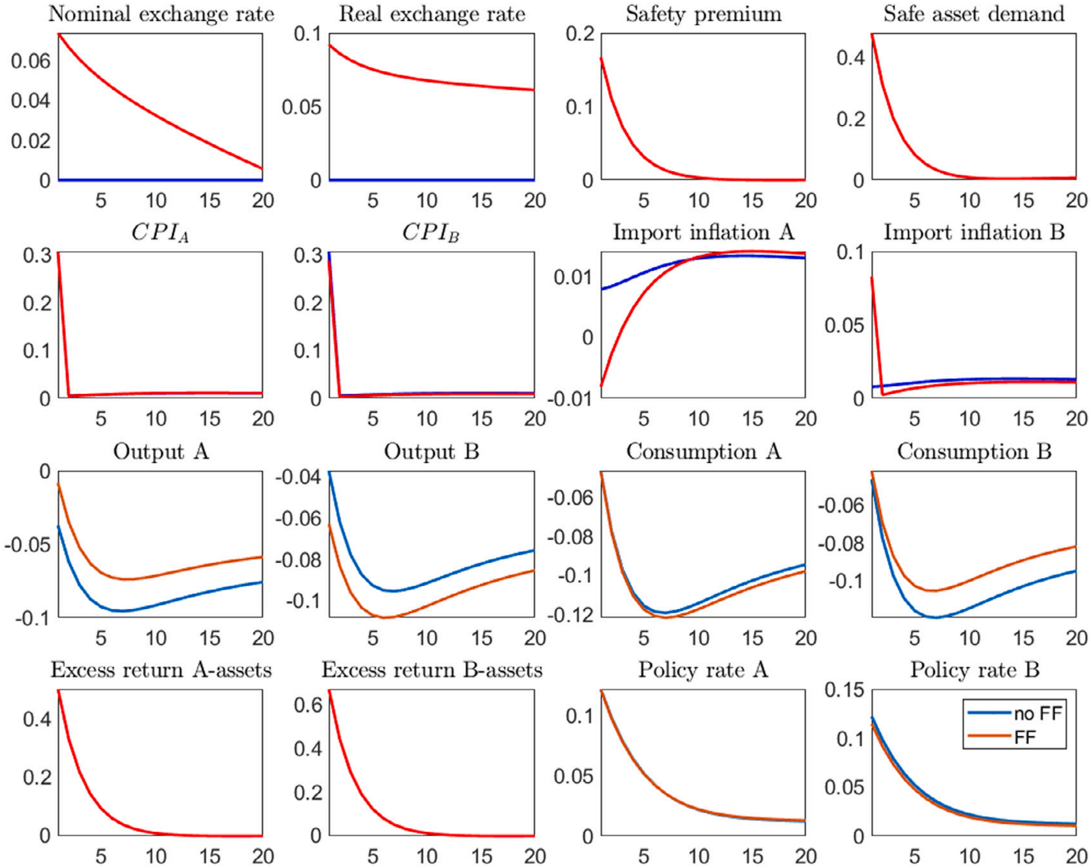

Fig. 6. Effects of an unanticipated symmetric increase of tariff rates in both countries by 2 percentage points. y-axis in percent. Red lines: baseline model. Blue lines: model without financial frictions. An increase in the nominal exchange rate denotes an appreciation of the safe asset currency. Safety premium: excess return of B-assets over A-assets. Safe Asset Demand: Increase in bank holdings of country-A assets relative to country-B assets. Import price inflation: depicted as pre-tariff. Excess returns: Expected return of domestic asset over the respective domestic deposit rate. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

rates in both countries perfectly co-move. The exchange rate, which in this setting follows the simple uncovered interest rate parity condition, Eq. (17), remains unchanged.

$$E_t\left[\frac{\mathcal{E}_{t+1}^{AB}}{\mathcal{E}_t^{AB}} R_t^B \Pi_{t+1}^B\right] = E_t[R_t^A \Pi_{t+1}^A]. \tag{17}$$

That is, the simple model cannot explain an appreciation of the US dollar in a scenario of a synchronized tariff rate increases.

The story is different in the full model (Fig. 6, red IRFs). Here, the responses of consumption, output and consumer prices resemble the dynamics in the simple model in the response to the tariff rate shock. They are also very similar in both countries. Importantly, however, they are not perfectly symmetric. For one, in the steady state, country A has a small trade deficit that balances its steady-state excess earnings on higher-yield foreign assets in the capital balance. This implies that the tariff rate hike lifts inflation in that country slightly more than in country B. In both countries, the rising inflation rates lead central banks to raise their policy rates and the real rates on deposits increase. Rising funding costs of banks reduce the operating value of banks. Hence, the incentive constraint (Eq. (3)) becomes more restrictive. Banks are forced to delever and reduce their asset holdings.[27] This lowers bond prices and raises the excess return on assets in both countries. As the incentive constraint binds more restrictively in terms of country-B bonds, banks shift their portfolio towards safer country-A assets and the safety premium that banks are willing to forego in order to hold safer bonds rises.

[27] Gertler and Karadi (2011) observe this deleveraging by financial intermediaries for an exogenous increase in the policy rate. The same holds for monetary policy shocks in our two-country model. We discuss the effects of monetary policy shocks in Appendix C.6.1. In addition, we discuss TFP shocks in Appendix C.6.2.

To see this, note that – from the viewpoint of banks in country A – Eq. (8) and (9) describe the marginal values of holding assets in terms of real spreads. In optimum, banks choose their portfolio considering the returns earned on their assets as well as the implications of their asset holdings for the tightness of the incentive constraint. Thus, the aforementioned equations, together with Eqs. (5) and (6), imply that

$$E_t[\Omega^A_{t+1}(\widetilde{R}^B_{b,t+1} - R^A_{d,t})] = \frac{\lambda_{BA}}{\lambda_{AA}} E_t[\Omega^A_{t+1}(R^A_{b,t+1} - R^A_{d,t})]. \tag{18}$$

As $\lambda_{BA} > \lambda_{AA}$, Eq. (18) implies that the increase in the excess return earned on country-A bonds must be met by a disproportionately large increase in the excess return earned on country-B bonds, i.e. an increase in the safety premium.

The nominal exchange rate in the model is determined by the portfolio decisions of banks in both countries, which have access to assets of both countries. Dividing both sides of Eq. (18) by $E_t[\Omega_{t+1}]$, adding $R^A_{d,t}$ and using $\widetilde{R}^B_{b,t} = \frac{RER_t}{RER_{t-1}} R^B_{b,t}$ and $\frac{RER_t}{RER_{t-1}} = \frac{\mathcal{E}^{A,B}_t}{\mathcal{E}^{A,B}_{t-1}} \frac{\Pi^B_t}{\Pi^A_t}$, yields an extended version of the UIP condition,

$$\frac{E_t[\mathcal{E}^{A,B}_{t+1}]}{\mathcal{E}^{A,B}_t} = \frac{\left(\frac{\lambda_{BA}}{\lambda_{AA}}(E_t[R^A_{b,t+1}\Pi^A_{t+1}] - R^A_{d,t}E_t[\Pi^A_{t+1}]) + R^A_{d,t}E_t[\Pi^A_{t+1}]\right)}{E_t[R^B_{b,t+1}\Pi^B_{t+1}]}, \tag{19}$$

where we omit covariance terms for the sake of providing intuition in a simplified setting.[28]

Expected real bond returns in their respective domestic currencies, $R^A_{b,t+1}$ and $R^B_{b,t+1}$, increase, as well as the real deposit rate, $R^A_{d,t}$ and the expected inflation rates. In sum, the result is only a small decrease of $\mathcal{E}^{A,B}_t$, i.e. an appreciation of the safe asset currency of country A. The rather mild appreciation of the safe asset currency after a synchronized tariff rate increase is in line with our empirical findings. In Appendix C.6.3, we show that this applies as well for a synchronized tariff rate shock that is anticipated four quarters in advance. In the same appendix, we show that a unilateral tariff rate increase by the safe asset country (i.e., the US), the exchange rate appreciation is more pronounced, but still smaller as in the case of a TPU shock.

3.3.2. The effect of trade policy uncertainty in the model without financial frictions

Trade policy uncertainty is captured in the model by stochastic volatility in the exogenous tariff rate process (Eq. (14)). An increase in trade tensions is represented by an increase in the standard deviation of tariff rates set by both countries. Motivated by the surge in TPU in 2018–19, and in line with Caldara et al. (2020), we consider a rise in the standard deviation of the tariff rate of 3 percentage points in our experiments.[29] Fig. 7 shows the effects of an increase in the full model (red lines) and the model without asymmetric financial friction (blue lines).

In the latter case, the effect of an increase of global trade policy uncertainty on the exchange rate is muted, as the responses in both countries are symmetric (despite trade invoicing, which plays a negligible role). Rising trade policy uncertainty induces a decrease in consumption by risk-averse households in both countries, due to a precautionary savings motive. Consequently, output declines as well. At the same time, price-setting firms raise their markups over their marginal costs due to their convex profit curve. Consumer price inflation, including import price inflation increases in both countries. As central banks in both countries in the simple model react by lifting their policy rates to the same extent, the exchange rate does not move.[30]

Two other observations are noteworthy. First, the effects of TPU on real activity remain small in this setting. This is in line with results from the literature on uncertainty shocks in DSGE models, which find that typically very large second-moment shocks are required to generate relevant effects on macroeconomic quantities (see, e.g. Born and Pfeifer, 2014; Bonciani and Roye, 2016; Basu and Bundick, 2017). Secondly, in both countries the central bank's response to firms' precautionary price markups raises the short-term real interest rate. This latter result carries over to the model with financial frictions.

3.3.3. The safe asset channel and the USD appreciation

Fig. 7 shows our key theoretical finding: that the inclusion of the asymmetric financial friction and the associated safe asset channel in our baseline model (red lines) tilts the response of the nominal exchange rate to an increase in TPU toward a substantial appreciation of country A's currency, i.e. the US dollar.

Note that the channels at work in the model without financial frictions are present in the full model as well. Whereas the sharp exchange rate movement drives a wedge between output, prices, and consumption of both countries, global aggregates have qualitatively similar dynamics as in the model without financial frictions, i.e. global consumption and output decline, whereas the precautionary markup motive by firms raises the average price level. Similarly, the average real short-term rate increases.

In the model with financial frictions, this implies that uncertainty regarding the path of tariff rates increases the banks' funding cost. The rising deposit rates present direct costs for banks and lower their net worth. Consequently, their continuation value drops

[28] Note that, if the financial friction was symmetric (i.e. $\lambda_{AA} = \lambda_{BA}$), Eq. (19) would collapse to $\frac{E_t[\mathcal{E}^{A,B}_{t+1}]}{\mathcal{E}^{A,B}_t} = \frac{E_t[R^A_{b,t+1}\Pi^A_{t+1}]}{E_t[R^B_{b,t+1}\Pi^B_{t+1}]}$, a version of Eq. (17) in terms of expected yields on banks' assets. In that case the key mechanism of the model is switched off. A version of Eq. (19) with the covariance terms, which are omitted here, is presented in Appendix C.4.

[29] In order to capture the effects of TPU shocks, we solve the model with a third-order approximation to equilibrium dynamics using the non-linear moving average by Lan and Meyer-Gohde (2013).

[30] TFP uncertainty shocks have similar effects in this model. For the case of the closed economy the dynamic consequences of these shocks have been discussed extensively in the literature. A good summary of potential effects of uncertainty in the closed-economy setting can be found in Born and Pfeifer (2014).

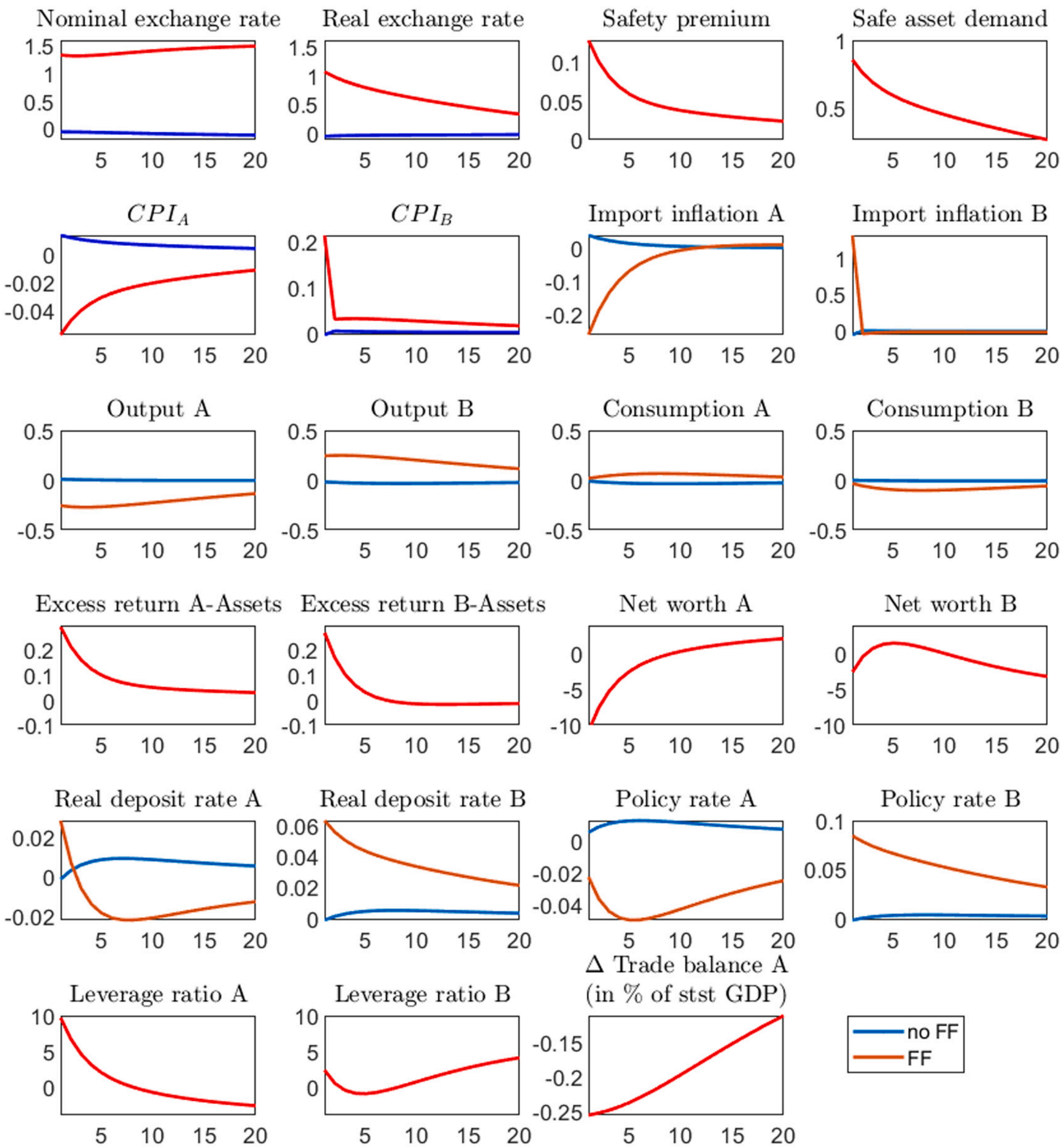

Fig. 7. Effects of a TPU shock. *y*-axis in percent. Red lines: baseline model. Blue lines: model without financial frictions. An increase in the nominal exchange rate denotes an appreciation of the safe asset currency. Safety premium: excess return of B-assets over A-assets. Safe Asset Demand: Increase in bank holdings of country-A assets relative to country-B assets. Import price inflation depicted as pre-tariff. Excess returns: Expected return of domestic asset over the respective domestic deposit rate. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

(via Eq. (2)) and depositors tighten the incentive constraint of banks (Eq. (3)), forcing them to delever and to sell assets.[31] As asset prices fall, the excess returns of bond yields over deposit rates increase. Because the constraint binds more tightly with regard to country-B assets, banks reduce their exposure to country-B assets in the process of deleveraging and shift their portfolio towards safer country-A assets. The safety premium rises. The net worth of banks in country A falls deeper than their country-B counterpart. This is due to the appreciation of the safe asset currency. The balance sheet of banks in country-B is in terms of its domestic currency. The appreciation of country-A currency supports the price of country-A bonds on the balance sheet of country B. Still, the net worth of country-B banks remains persistently subdued due to the elevated real deposit rate, which results from the central bank's reaction to inflationary pressures.

[31] Similarly, Mikkelsen and Poeschl (2022) show in a model with banks à la (Gertler and Karadi, 2011) that following an uncertainty shock, their decreasing continuation value forces banks to delever. The authors dub this the financial constraint channel. Dlugoszek (2019) documents this in a similar model.

Higher uncertainty regarding future tariff rate changes also translates directly into additional uncertainty regarding banks' future bond portfolio, since trade balance fluctuations have to be balanced by net foreign asset adjustments between countries (see Eq. (16)). Put differently, banks face uncertainty regarding changing needs to adjust their portfolio in order to meet potential future swings in the trade balance triggered by tariff rate changes.

In the face of higher trade policy uncertainty, risk-averse banks place a lower value on holding long-term bonds. This is due to the negative covariance between the banks' stochastic discount factor (SDF) and excess bond returns conditional on trade policy uncertainty shocks, which generates a risk premium. To see this, consider, for the case of US banks, Eq. (5), which is restated here for convenience (without the subscript j due to symmetry of banks):

$$\begin{aligned} v_{bt}^{AA} &= \beta E_t[\Omega_{t+1}^A(R_{b,t+1}^A - R_{d,t}^A)] \\ &= \beta E_t[\Omega_{t+1}^A](E_t[R_{b,t+1}^A] - R_{d,t}^A) + Cov(\Omega_t^A; R_{b,t}^A). \end{aligned}$$

It states that the marginal value of additional holdings of asset A for country-A banks, v_{bt}^{AA}, equals the expected product of the banks' SDF, $\Omega_{t+1}^A \equiv \Lambda_{t,t+1}^A((1-\theta) + \theta(1+\mu_{t+1}^A)v_{n,t+1}^A)$ and expected excess return earned on the asset $(R_{b,t+1}^A - R_{d,t}^A)$. In both countries, banks' SDFs increase with higher funding costs as the latter raise the marginal value of an additional unit of net worth, (e.g., for country A: v_{nt}^A depends positively on $R_{d,t}^A$ via Eq. (10)) and tighten the incentive constraint.[32] At the same time, the downward pressure on net worth reduces banks' demand for A-assets via Eq. (7) and lowers the price on A-bonds on impact. Thus, their return also falls sharply on impact (see Eq. (1)). After their initial decline, the bond price recovers and returns are positive in the subsequent periods. The latter is reflected by increasing expected excess return on bonds over deposits in the period after the TPU shock (see Fig. 7, which displays excess returns on assets over the respective domestic rate). Due to the initial decline of the bond return, the covariance between the banks' SDF and the bond's return conditional on trade policy uncertainty shocks is negative. The negative covariance causes the banks valuation of assets, $v_{b,t}^{AA}$, to drop, reducing the price of A-assets. Similarly, the marginal value of holding B-assets for A-banks, v_t^{AB}, and the marginal values for bond holdings for B-banks, v_t^{BA}, and v_t^{BB}, drop and with them bond prices (for further details see Appendix C.3). This channel amplifies the deleveraging of banks and their portfolio shift towards safer assets, i.e. assets for which the balance sheet constraint is less binding.

In Appendix C.5.5, we illustrate that, the higher the steady-state leverage ratio, the more sensitive banks' net worth is to rising costs of external financing and to fluctuations in asset prices, the more volatile asset returns are, and the higher the safety premium is in response to rising trade policy uncertainty. Thus, also the effect on the exchange rate is amplified for TPU shocks.

Whereas the effects of TPU shocks and tariff rate shocks on financial variables are qualitatively similar, the effect on the exchange rate is stronger for TPU shocks. The symmetric tariff rate shock has a direct and substantial first-order effect on inflation rates in both countries, both of which are driven up by the shock. This determines the response of the exchange rate. In contrast, TPU shocks do not have this effect. They drive country-specific inflation rates in opposite directions.[33]

The dynamic response of non-financial variables to TPU shocks in the model is dominated by the exchange rate channel. Fig. 7 shows that the appreciation of the currency of country A raises imports price inflation in country B on impact. The swiftness of this increase is due to producer currency pricing by country-A exporters. Conversely, as country-B exporters apply local currency pricing, the decline in import price inflation in country A is more gradual and in its pace of adjustment governed by Calvo price stickiness. These import price dynamics are reflected by the responses of consumer price inflation in both countries. Importantly, accounting for dollar dominance in trade implies a persistent wedge between CPI inflation in both countries following the increase in trade policy uncertainty, which amplifies the exchange rate response. In Appendix C.5.3, we discuss the robustness of the main results to alternative pricing assumptions. We show that if we ignore dollar dominance and let exporters in both countries apply producer currency pricing, the CPI wedge is less persistent and the appreciation of safe asset currency is reduced in magnitude.

The exchange rate channel also dominates the dynamics of production and consumption in both countries. With the appreciation of the safer currency, country-A products become less competitive and country-A exports and output decline. The imports from country B allow households in country A to increase consumption. Conversely, in country B output rises and consumption falls. The real trade deficit of Country A thus widens. The magnitude of country-specific output and consumption responses is substantial due to the exchange rate movements. With the increase of imports and the decrease of exports the trade deficit of country A expands markedly as a response to the TPU shock in the model. The widening of the trade deficit is in line with our empirical results and implies that an increase in trade policy uncertainty may counteract US trade policy attempts to increase US competitiveness. Whereas the increase in country-A consumption is consistent with higher US consumption after a positive empirical TPU shock, the country-A output response in the model is at odds with the increase in US GDP in response to the shock. The latter suggests that by abstracting from other expenditure components – such as, e.g., inventories that might rise due to precautionary stockpiling – our model overestimates the effect of expenditure switching on overall US GDP. We discuss this in more detail in Section 3.5.

[32] With the divergence of consumption growth between countries, the SDF of households diverge as well. However, banks' SDFs both rise as they are dominated by the elevated marginal value of net worth in both countries (see, Appendix C.3).

[33] In addition, trade policy uncertainty shocks directly affects higher-order terms contributing to the appreciation of country A's currency. Appendix C.4 provides a version of the expanded UIP conditions that spells out the covariances of the expected variables. Contrary, higher-order terms play no role in the transmission of tariff rate shocks in the linear model and only a negligible one when a third-order approximation to equilibrium dynamics is applied (see, Appendix C.6.3).

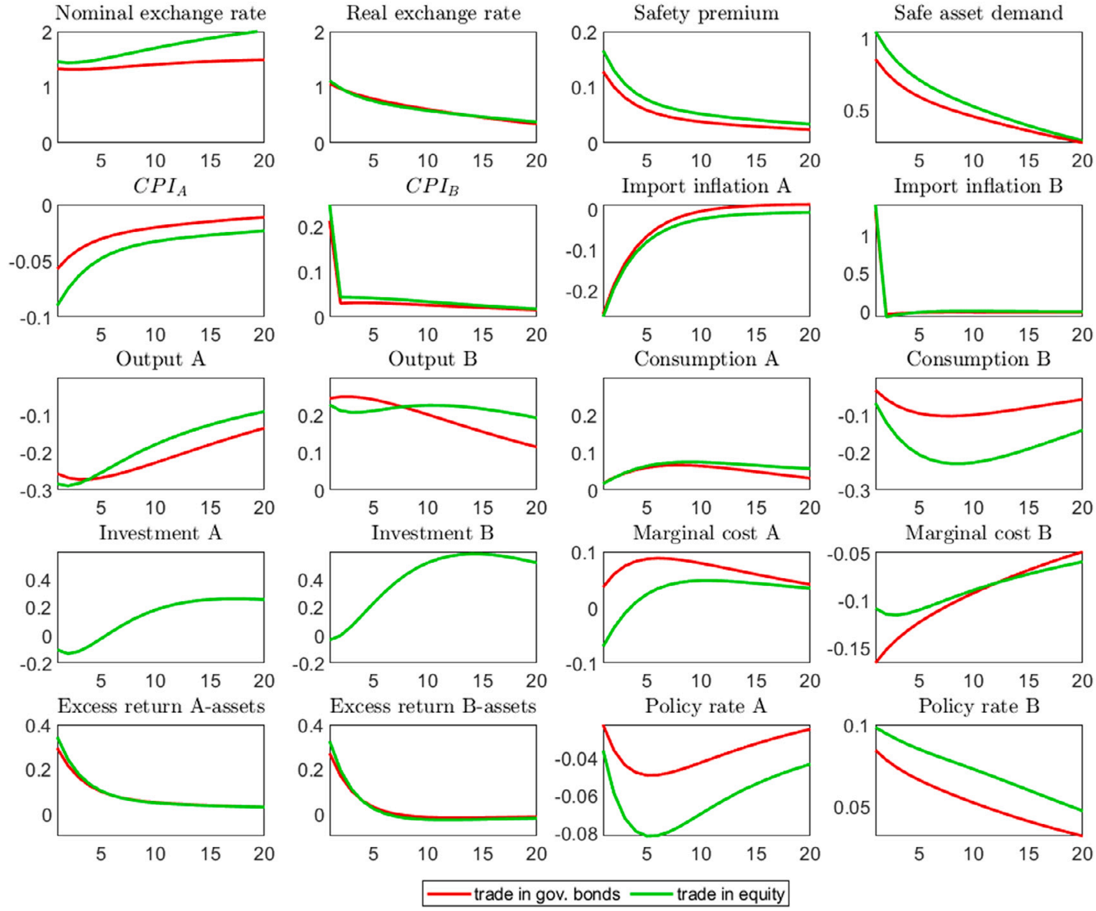

Fig. 8. Effects of a TPU shock. y-axis in percent. Red lines: Model with government bonds as internationally traded assets. Green lines: Model with claims on capital stock as internationally traded assets. An increase in the nominal exchange rate denotes an appreciation of the safe asset currency. Safety premium: excess return of B-assets over A-assets. Safe Asset Demand: Increase in bank holdings of country-A assets relative to country-B assets. Import price inflation: depicted as pre-tariff. Excess returns: Expected return of domestic asset over the respective domestic deposit rate. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

3.3.4. *A model version with capital in production and international asset trade*

So far, we assumed that international asset trade takes place in long-term government bonds. Here we consider a version of the model that features capital in production. Asset trade takes place in claims on the private capital stock, or equity, instead of government bonds. This adds another channel through which trade policy uncertainty affects equilibrium dynamics. Now the portfolio choice by banks feeds back to the production side of the economy.

The production function of firms now reads $Y_{p,t}^A = A_t^A (K_t^A)^\alpha (L_t^A)^{1-\alpha}$, where K_t^A, L_t^A, and A_t^A denote capital input, labor input, and TFP, respectively. α is the output elasticity with respect to capital goods. The production of capital is subject to investment adjustment cost. The real return on capital, $R_{k,t}^A = \frac{P_{m,t}^A \alpha \frac{Y_t^A}{K_{t-1}^A} + (1-\delta) Q_t^A}{Q_{t-1}^A}$, depends on the marginal product of capital and the resale value of capital, Q_t^A. The output price of producing firms, $P_{m,t}^A$, represents the marginal cost for retailers, as in Gertler and Karadi (2011) and δ is the depreciation rate. For more details on the model, we refer to Appendix C.5.2. In this exercise, we calibrate the additional parameters to the values chosen in Caldara et al. (2020). Hence, $\alpha = 0.36$, $\delta = 0.025$ and $\phi_K = 10$.

Fig. 8 shows that, given the calibration, the effects of a TPU shock are largely similar to those in the baseline model. In particular, the rising return of risky assets and the associated increase in the spread over the deposit rates are barely affected by the type of assets traded. Thus, the safety premium on assets issued in country A and the increase in safe asset demand are present in both models, and the appreciation of the nominal exchange rate of the safe currency is similar. Overall, the key mechanism that links TPU and the nominal exchange rate remains the same in this setup. Nonetheless, the presence of capital in production introduces an additional dimension to the transmission of the shock.

Global investment is reduced on impact as falling asset prices lead to a contraction of banks' balance sheets. However, globally, this effect is quickly reversed. The subsequent increase in investment is consistent with the response of capital that Caldara et al.

(2020) document for the case of separable preferences. They attribute this to an increase in the labor supply due to a negative wealth effect, which also raises increase of capital in production. In our case the negative wealth effect due to falling consumption holds in the global aggregate.[34] Additionally, in the model investment in country B is supported by higher demand for output goods by customers in country A due to the appreciation of its currency.[35] Despite these differences to our baseline case, the model remains able to explain the safe asset currency appreciation.

3.4. The robustness of the safe asset channel

In this section, we briefly summarize how the implications of higher trade policy uncertainty for the USD exchange rate are affected by various alternative modeling choices. The reader is referred to Appendix C.5 for further detail. It turns out that the effect of trade policy uncertainty shocks is remarkably robust.

Trade policy uncertainty in a model with global value chains In Appendix C.5.1, we show that the link between trade policy uncertainty and the appreciation of the safe asset currency remains intact, when we expand the model to feature global value chains. In that setting, foreign and domestic intermediate goods are used as an input in the production of final goods. Both types of goods are internationally traded and can be subject to tariffs. In that model, exchange rate movements are attenuated. This is not surprising, as tariff rate changes of a given size have smaller effects on the trade balance. This reduces the impact of uncertainty regarding future tariff rate changes. To see this, consider e.g. an increase in the tariff rate on imported final goods. This lets consumers turn to domestically produced final goods. With the ensuing expansion of domestic production, the demand for foreign intermediates rises and attenuates the effect of tariffs on the trade balance. Conversely, a tariff on imported intermediates raises domestic firms' production costs making their goods less competitive on the international market; final goods exports decrease. Thus, global value chain participation can dampen shifts in the trade balance and thus exchange rate movements. With the effects of tariff rate changes reduced, the impact of rising trade policy uncertainty is reduced as well. Its effect on the appreciation of the safe asset currency remains sizable, however.

Furthermore, we show that an increase in uncertainty regarding tariffs on final goods has a far stronger effect than uncertainty regarding tariffs on intermediate goods in the model. Final good inflation features both the SDF of households, which determines the valuation of future consumption and income streams of agents and is key for generating effects of uncertainty shocks. Final good inflation is relevant for determining nominal interest rates and directly features the expanded UIP condition that follows from banks' optimization problem.

Alternative assumptions on pricing Firms respond to uncertainty by setting precautionary markups. Thus, our assumptions on price setting may, in principle, affect our results. Inspired by dollar dominance in global trade invoicing, we assume producer currency pricing (PCP) by exporters in country A and local currency pricing (LCP) by exporters in country B. In addition, we opt for Calvo pricing with an average price duration of one year. In Appendix C.5.3, we show that the safe asset channel is robust to the modeling assumptions of symmetric PCP, Rotemberg pricing or a flatter Phillips Curve.

Asymmetric country sizes The baseline model features two equally-sized countries. In Appendix C.5.4, we show that the effect of TPU shocks on the exchange rate remains intact when we adjust the size of the safe asset country A to match the weight of the US in the world economy. Country A now makes up a sixth of the world economy, while "the rest of the world" (i.e., country B) has a weight of five sixths in the world economy. In that setting, trade and hence trade policy uncertainty becomes less relevant for country B. Its prices move less in response to the TPU shock, and the CPI differential between countries shrinks. This reduces the amplification of exchange rate changes via relative price movements. Whereas the safe asset channel remains active, the appreciation of the safe asset currency is smaller, albeit still sizable. However, that setting with "the rest of the world" being a bloc with frictionless internal trade downplays the importance of trade policy uncertainty for countries outside of the US, which in reality share similar uncertainty regarding trade frictions.

3.5. Discussion of the effects of trade policy in the model compared to the data

Overall, our baseline model is able to explain key results of our empirical investigation. With a plausible calibration, trade policy uncertainty and tariff rate shocks both trigger an appreciation of the safe asset currency, i.e., the US dollar. Importantly, the response of the exchange rate to increases in TPU is more pronounced. The dynamics of financial variables such as the safe asset premium and the increase in the relative demand for safe assets in the model qualitatively match the increase in the US Treasury premium and the rising foreign holdings of US long-term securities. This close alignment validates the role of the safe asset channel for the propagation of TPU shocks.

In addition, our model rationalizes the decline in country A's import and consumer price inflation in the SVAR, as well as the increase in US export prices (in foreign currency) and foreign CPI. Also the widening of the trade deficit and the increase in country-A consumption do align qualitatively with the widening of the US trade deficit and the increase in US consumption in response to the rise in TPU in the data.

[34] In addition, uncertainty can also increase the demand for capital via other channels discussed in the literature (cf. Born and Pfeifer 2014).

[35] The finding that an increase in uncertainty can raise investment despite falling capital prices has been shown by Dlugoszek (2019) for the case of a closed economy with banks as in Gertler and Karadi (2011) and a TFP uncertainty shock.

With regards to production in both countries, however, our model does not perform equally well and is at odds with the responses of US and global industrial production in the SVAR. This suggests that our stylized model overestimates the link between exchange rate movements and production. Moreover, our model abstracts from other aspects that potentially shape the empirical findings. We abstract from modeling global commodity prices – including global oil prices – that are dampened markedly in the data and thus likely support US output. We also do not model inventories — a channel likely relevant in the data (see Caglayan et al. 2012). Moreover, under dominant currency pricing, a USD appreciation hurts global trade by dampening US-invoiced trade between country pairs that do not involve the US itself (see Gopinath et al. 2020). Our simplified two-country model cannot capture such effects on global production.

4. Conclusion

We show that the US dollar response to trade policy uncertainty (TPU) shocks is key to assessing the overall impact of trade policies. Using SVAR models, we find that a sudden increase in TPU persistently strengthens the USD against a broad currency basket. Such shocks account for a sizable fraction of the USD appreciation between 2018 and 2019. We trace out the channels underlying this finding in an open economy New Keynesian model featuring financial frictions in banks' asset holdings and a different pledgeability of US and non-US assets that gives rise to a safety premium for US assets. In this setting, TPU shocks trigger an increase in the relative demand for safer US assets, thereby causing an appreciation of the US dollar.

Our empirical and theoretical results suggest that historical US trade policy uncertainty shocks (such as in 2018–19) moved the USD to a larger extent than US tariff rate shocks. From the US perspective, the resulting decline in US import prices and increase in (foreign currency) US export prices reduces competitiveness markedly, which offsets to a large extent competitiveness gains due to higher import tariffs.

While we focus on the effect of trade policy on the US dollar, the link between higher TPU and a nominal exchange rate appreciation likely generalizes to other currencies that are considered a safe haven.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Replication package for 'US trade policy and the US dollar' (Original data) (Mendeley Data).

Appendix A. Supplementary data

Supplementary material related to this article can be found online at https://doi.org/10.1016/j.jinteco.2024.103970.

References

Baker, S.R., Bloom, N., Davis, S.J., 2016. Measuring economic policy uncertainty. Q. J. Econ. 131 (4), 1593–1636.
Basu, S., Bundick, B., 2017. Uncertainty Shocks in a Model of Effective Demand. Econometrica 85, 937–958.
Bhattarai, S., Chatterjee, A., Park, W.Y., 2020. Global spillover effects of US uncertainty. J. Monetary Econ. 114, 71–89.
Board of Governors of the Federal Reserve System, 2022. Financial Stability Report (November 2022).
Boehl, G., Strobel, F., 2024. Estimation of DSGE models with the effective lower bound. J. Econom. Dynam. Control 158, 104784.
Bonciani, D., Roye, B.v., 2016. Uncertainty shocks, banking frictions and economic activity. J. Econom. Dynam. Control 73 (C), 200–219.
Born, B., Pfeifer, J., 2014. Policy risk and the business cycle. J. Monetary Econ. 68, 68–85.
Bu, C., Rogers, J., Wu, W., 2021. A unified measure of Fed monetary policy shocks. J. Monetary Econ. 118, 331–349.
Caglayan, M., Maioli, S., Mateut, S., 2012. Inventories, sales uncertainty, and financial strength. J. Bank. Financ. 36 (9), 2512–2521.
Caldara, D., Iacoviello, M., Molligo, P., Prestipino, A., Raffo, A., 2019. Does Trade Policy Uncertainty Affect Global Economic Activity? FEDS Notes. Board Governors Federal Reserve Syst..
Caldara, D., Iacoviello, M., Molligo, P., Prestipino, A., Raffo, A., 2020. The economic effects of trade policy uncertainty. J. Monetary Econ. 109, 38–59.
Calvo, G.A., 1983. Staggered prices in a utility-maximizing framework. J. Monetary Econ. 12 (3), 383–398.
Coeurdacier, N., Rey, H., 2013. Home bias in open economy financial macroeconomics. J. Econ. Lit. 51 (1), 63–115.
Deutsche Bundesbank, 2020. Consequences of increasing protectionism. Monthly Report January 2020, 45–66.
Dieppe, A., van Roye, B., Legrand, R., 2018. The Bayesian estimation, analysis and regression toolbox (BEAR) – version 4.2 – European Central Bank.
Dlugoszek, G., 2019. Essays on macroeconomic consequences of uncertainty (Ph.D. thesis). Humboldt-Universität zu Berlin, Wirtschaftswissenschaftliche Fakultät.
Du, W., Im, J., Schreger, J., 2018. The U.S. treasury premium. J. Int. Econ. 112, 167–181.
Du, W., Schreger, J., 2016. Local currency sovereign risk. J. Finance 71/3, 1027–1070.
Furceri, D., Hannan, S.A., Ostry, J.D., Rose, A.K., 2019. Macroeconomic Consequences of Tariffs. IMF Working Papers 2019/009, International Monetary Fund.
Gabaix, X., Maggiori, M., 2015. International liquidity and exchange rate dynamics. Q. J. Econ. 130 (3), 1369–1420.
Georgiadis, G., Müller, G.J., Schumann, B., 2024. Global risk and the dollar. J. Monetary Econ. 103549.
Gertler, M., Karadi, P., 2011. A model of unconventional monetary policy. J. Monetary Econ. 58 (1), 17–34.
Gertler, M., Karadi, P., 2013. QE 1 vs. 2 vs. 3. . . : A framework for analyzing large-scale asset purchases as a monetary policy tool. Int. J. Central Bank. 9 (1), 5–53.
Ghironi, F., Ozhan, G.K., 2020. Interest rate uncertainty as a policy tool. National Bureau of Economic Research Working Paper, No. 27084.
Gopinath, G., Boz, E., Casas, C., Díez, F.J., Gourinchas, P.-O., Plagborg-Møller, M., 2020. Dominant currency paradigm. Amer. Econ. Rev. 110 (3), 677–719.

He, Z., Krishnamurthy, A., Milbradt, K., 2019. A model of safe asset determination. Amer. Econ. Rev. 109 (4), 1230–1262.
Itskhoki, O., Mukhin, D., 2021. Exchange rate disconnect in general equilibrium. J. Polit. Econ. 129 (8), 2183–2232.
Jeanne, O., Son, J., 2024. To what extent are tariffs offset by exchange rates? J. Int. Money Finance 142, 103015.
Jiang, Z., Krishnamurthy, A., Lustig, H., 2021. Foreign safe asset demand and the dollar exchange rate. J. Finance 76 (3), 1049–1089.
Jiang, Z., Krishnamurthy, A., Lustig, H., 2023. Dollar Safety and the Global Financial Cycle. Rev. Econ. Stud..
Jurado, K., Ludvigson, S.C., Ng, S., 2015. Measuring uncertainty. Am. Econ. Rev. 105 (3), 1177–1216.
Kulish, M., Morley, J., Robinson, T., 2017. Estimating DSGE models with zero interest rate policy. J. Monetary Econ. 88, 35–49.
Lan, H., Meyer-Gohde, A., 2013. Solving DSGE models with a nonlinear moving average. J. Econ. Dyn. Control 37 (12), 2643–2667.
Lindé, J., Pescatori, A., 2019. The macroeconomic effects of trade tariffs: Revisiting the lerner symmetry result. J. Int. Money Finance 95, 52–69.
Meeks, R., Nelson, B., Alessandri, P., 2017. Shadow banks and macroeconomic instability. J. Money Credit Bank. 49 (7), 1483–1516.
Mikkelsen, J., Poeschl, J., 2022. Banking Panic Risk and Macroeconomic Uncertainty. Mimeo.
Müller, G., Wolf, M., Hettig, T., 2024. Delayed Overshooting: The Case for Information Rigidities. Am. Econ. J. Macroecon. 16 (3), 310–342.
Smets, F., Wouters, R., 2007. Shocks and frictions in US business cycles: A Bayesian DSGE approach. Amer. Econ. Rev. 97 (3), 586–606.
Trani, T., 2015. Asset pledgeability and international transmission of financial shocks. J. Int. Money Finance 50 (C), 49–77.
Woodford, M., 1998. Public debt and the price level. Mimeo.
Woodford, M., 2001. Fiscal requirements of price stability. J. Money Credit Bank. 33, 669–728.